

ANALYTIC CONSTRUCTION OF RATIONAL CURVES ON FANO MANIFOLDS

YUN-HENG DU, BIN GUO, AND SONG-YAN XIE

ABSTRACT. Inspired by constructions of entire curves in Oka geometry, we construct rational curves on complex Fano manifolds by holomorphic disk surgeries. Positive Ricci curvature guides an area-decreasing deformation, and an affine-lift perturbation enlarges the source. Explicit estimates control the area and weighted derivative changes of both operations. A finite hierarchy of principal-map recoveries then gives a sphere meeting a prescribed compact fiber without being contained in it, with a sharp metric area bound. The construction uses analytic compactness to pass from the disks to a sphere. Combined with classical algebraic arguments, it gives new proofs in characteristic zero of two known results: rational connectedness of Fano manifolds, due to Campana and Kollár–Miyazaki–Mori, and Mori’s solution of Hartshorne’s conjecture on ample tangent bundles.

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1. INTRODUCTION AND MAIN RESULTS

1.1. The analytic construction problem

Our aim is to construct rational curves on complex Fano manifolds by explicit analytic operations, with control of the intermediate disks. A Fano manifold X is connected, smooth, and compact, with ample anticanonical line bundle K_X^{-1} . In his monograph *Ample subvarieties of algebraic varieties*, Hartshorne [27] asked whether a smooth complex projective variety with ample tangent bundle must be projective space. Mori’s solution [39] made the construction of rational curves a central tool of higher-dimensional algebraic geometry. His construction combines deformation theory and bend-and-break, which also apply in characteristic zero, with Frobenius in positive characteristic and specialization back to $\mathbb{C}$.

The search for an analytic construction of rational curves has a long history. In 1980, Siu and Yau [45] used energy-minimizing harmonic spheres in their proof of the Frankel conjecture. Their

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argument uses positive holomorphic bisectional curvature, a stronger condition than the Ricci positivity that Yau’s prescribed-Ricci theorem provides on every Fano manifold [48]. Already in 1982, Yau asked whether any two points of a compact Kähler manifold with positive Ricci curvature could be joined by a chain of rational curves [49, p. 44]. Rational connectedness of Fano manifolds was subsequently established by Campana and by Kollár, Miyaoka, and Mori [6, 35]. The question of constructing the curves by analytic methods remained a separate challenge. Siu later proposed approaching this problem through complex Monge–Ampère equations, without reduction to positive characteristic [44].

The folklore conjecture that every Fano manifold is Oka provides another motivation; see [20, p. 660] for this open problem. For the general framework of Oka theory and its relation to Stein geometry, we refer to Forstnerič’s monograph [19]. Alarcón and Forstnerič opened a new direction in Oka geometry by introducing Oka–1 manifolds and conjecturing, in particular, that every rationally connected projective manifold is Oka–1 [1, Conjecture 9.1]. Our recent proof of the projective statement [14, Corollary 1.8] brought new insights and methods for constructing entire curves. Guo and Xie had already suggested in [25, Section 5] that studying entire curves with slow growth could lead to analytic constructions of entire and rational curves on Fano manifolds. Our companion results motivate this approach, but the proof below neither invokes them nor assumes that the target is Oka–1.

Following the direction suggested by Guo and Xie [25], we pursue a precise growth condition: finite total area. The argument combines the Brody normalization and expanding-disk viewpoint of complex hyperbolicity with deformations of holomorphic maps from Oka geometry [46] and the compactness and removal principles for finite-area curves familiar from symplectic geometry. Positive Ricci curvature guides the deformations, keeping area uniformly bounded as the source radii tend to infinity. A nonconstant entire limit then extends across infinity to a rational curve; a suitable sphere may also appear earlier by bubbling.

A traditional Chinese pastry-making technique suggests a picture of the construction: a baozi is formed by gathering a disk of dough around its filling and closing the folds. Here the gathering recalls the deformation of holomorphic disks, while the closure recalls finite-area removal at infinity. Positive Ricci curvature provides the area control needed for this limiting step.

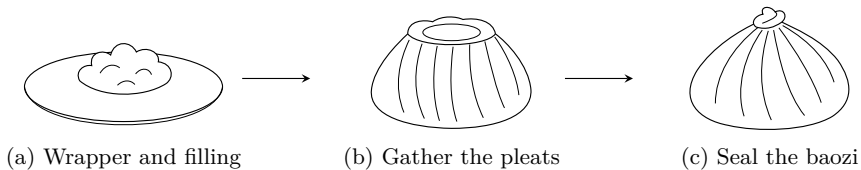


FIGURE 1. Shaping a baozi by gathering a dough wrapper around a filling and sealing the pleats. This traditional Chinese craft suggests an analogy for closure. In the proof, a finite-area entire principal map extends across infinity; bubbling is treated separately by marked compactness.

The first obstacle is to control area as the source radii tend to infinity: holomorphic approximation alone supplies no uniform bound. Finite total area is what allows the limiting map to extend across infinity; see Theorem 2.4.

Position can be lost through reparametrization. Classical Brody reparametrization moves the source centers [5], so even disks through a fixed point need not yield an entire curve through that point. Duval addressed this loss of location by replacing derivative selection with control of where area accumulates [15]; see also [16]. For a compact set charged by an Ahlfors current, his theorem produces an entire curve whose intersection with that set has positive area. This is an

important inspiration for our construction. In our finite-area setting, we retain position through a base constraint and two marked values, rather than invoke the Ahlfors-current theorem itself.

Even when the source center is fixed, bubbling can separate its prescribed local data from the nonconstant sphere components. The analysis of bubbling goes back to Sacks and Uhlenbeck's work on harmonic maps [43] and Gromov's theory of pseudoholomorphic curves [24]. Area may also concentrate near the source boundary. We must therefore obtain a nonconstant sphere satisfying both the required incidence conditions and the area bound.

Our construction addresses area control and localization together. We first use Ricci positivity to decrease area above an explicit threshold. The key is a Ricci-trace estimate, realized through approximation and interpolation by holomorphic families that preserve a scalar derivative condition and satisfy uniform estimates. To retain the geometric information under compactness, we use a relative formulation: the center may move in a compact fiber, while a derivative of the projection to the base remains normalized. This yields a sphere meeting but not contained in that fiber, with a sharp area bound; the pointwise specialization gives a rational curve through every prescribed point. The intermediate maps are given by holomorphic perturbation formulas; area and weighted derivative estimates control their costs, and recovery levels account for bubbling. The curves are constructed analytically over $\mathbb{C}$; the rational-connectedness and classification applications then use classical rational-quotient, chain-smoothing, and minimal-curve arguments in characteristic zero. We explain the mechanism after stating the main results.

1.2. Main results

To prove rational connectedness by our method, we need more than a curve through a point: we need one that escapes a fiber of the rational quotient. Our main result therefore produces a holomorphic sphere meeting a compact fiber without being contained in it. We use the Ricci and area conventions in (1). The fiber in the statement is the full inverse image in the specified neighborhood.

Theorem 1.1. *Let (X, ω) be a compact Kähler n -fold satisfying*

$$\mathrm{Ric}(\omega) \geq \kappa \omega, \quad \kappa > 0.$$

For an open subset $N \subset X$ and a complex manifold Y of positive dimension, consider a holomorphic map $q : N \rightarrow Y$ and a point $y_0 \in Y$. Suppose that the full fiber $F := q^{-1}(y_0)$ is nonempty and compact and that q is a submersion on a neighborhood of F in N . Then there is a nonconstant holomorphic map $u : \mathbb{P}^1 \rightarrow X$ such that

$$u(\mathbb{P}^1) \cap F \neq \emptyset, \quad u(\mathbb{P}^1) \not\subset F, \quad \int_{\mathbb{P}^1} u^* \omega \leq \frac{2\pi(n+1)}{\kappa}.$$

Yau's prescribed-Ricci theorem supplies a metric satisfying this hypothesis on every Fano manifold [48]; see Proposition 3.1. No Kähler–Einstein hypothesis is required. Conversely, the Ricci lower bound makes K_X^{-1} positive and hence ample, so the target is projective and the reduced image of the sphere is a rational curve; see Remark 3.2. The coefficient $2\pi(n+1)$ is sharp as a universal metric bound, as the projective-space example in Remark 2.12 shows. The relative viewpoint has a classical antecedent in Campana's generic-fiber escape criterion and his Fano theorem [6, Proposition 2.7 and Theorem 3.1]. Here we work near a specified compact full fiber of a holomorphic submersion and obtain a metric area bound.

Allowing the center to move within F lets the normalized base derivative pass to the principal limit whenever first-exit compactness has not already produced a sphere meeting but not contained in F . The final passage to the sharp area bound retains intersection with F and noncontainment in F , but need not preserve this derivative normalization. No tangent direction or contact order is prescribed.

Taking a coordinate map near a point gives the pointwise form immediately.

Theorem 1.2. *Let (X, ω) be a positive-dimensional compact Kähler manifold of complex dimension n , and suppose $\text{Ric}(\omega) \geq \kappa \omega$ for some $\kappa > 0$. Then, for every $x \in X$, there is a nonconstant holomorphic map $u_x : \mathbb{P}^1 \rightarrow X$ such that*

$$x \in u_x(\mathbb{P}^1), \quad \int_{\mathbb{P}^1} u_x^* \omega \leq \frac{2\pi(n+1)}{\kappa}.$$

Classical bend-and-break produces a rational curve C through a prescribed point of a Fano n -fold with

$$0 < -K_X \cdot C \leq n+1.$$

Mori's account [40, p. 749] attributes this form of the result to Kollár, by a technique similar to that of [39]. Together with $\kappa \omega \leq \text{Ric}(\omega)$, this degree bound gives the same pointwise area estimate. The contribution here is an analytic construction of the curve with direct control of its area. See [11, Chapter 3] for the classical argument.

Starting from our analytic construction, we give proofs of the following two classical theorems in characteristic zero, without reduction to positive characteristic. Recall that a smooth projective variety is rationally connected if two general points lie on the image of one morphism $\mathbb{P}^1 \rightarrow X$. Fujiki formulated this notion for compact complex spaces [21, Definition 2.8 and the following remark]. The following theorem was proved by Campana and by Kollár, Miyaoka, and Mori [6, 35].

Theorem 1.3. *Every positive-dimensional complex Fano manifold is rationally connected.*

Campana's criterion [6, Proposition 2.7] reduces rational chain connectedness to the existence of a rational curve meeting but escaping a general fiber of every quasifibration with positive-dimensional base. Theorem 1.1 supplies precisely this curve, and characteristic-zero smoothing of rational chains gives rational connectedness.

The second application is Mori's theorem [39], which resolves Hartshorne's ample-tangent conjecture stated above.

Theorem 1.4. *Let X be a connected smooth complex projective variety of dimension $n \geq 1$. If T_X is ample, then $X \simeq \mathbb{P}^n$.*

If T_X is ample, then $K_X^{-1} = \det T_X$ is ample, so X is Fano. The pointwise construction supplies the first rational curve. Classical two-point degree reduction and Mori's minimal-curve classification then apply over $\mathbb{C}$; Section 7 checks their hypotheses and gives precise references.

1.3. The surgeries and their quantitative control

For a holomorphic map $f : \mathbb{D}_R \rightarrow X$, write

$$\text{Area}_R(f) = \int_{\mathbb{D}_R} f^* \omega, \quad B_R(f) = \sup_{z \in \mathbb{D}_R} (R - |z|) |f_z(z)|_{\omega}.$$

The second quantity measures derivative concentration on the scale of the distance to the source boundary, as in Brody's reparametrization argument [5]. We use it both to choose uniform surgery parameters and to detect the sequences from which a bubble can be removed.

The first surgery fixes the center and one scalar derivative. In a holomorphic frame with unitary boundary values, one quadratic and $n-1$ linear multiplier components impose these constraints at Dirichlet cost $\pi(n+1)$. Their paired second-variation trace is

$$4\pi(n+1) - 2 \int_{\mathbb{D}_R} f^* \text{Ric}(\omega) \leq 4\pi(n+1) - 2\kappa \text{Area}_R(f).$$

This gives the threshold $A_* = 2\pi(n+1)/\kappa$. The multiplier attains the least cost under its interpolation and boundary conditions; see the calculation after (34). A projective lift, a corona theorem and a holomorphic retraction realize the trace by actual families with a uniform parameter radius and cubic area remainder. Selecting a real Hessian direction and the sign of the first derivative then gives a definite area decrease above the threshold.

The second surgery is particularly simple in the lifted coordinates. For a lift H and its projection Π , set

$$h_t(z) = \Pi(H(z) + tzH'(0)), \quad G_t(z) = h_t(z/(1+t)).$$

The first formula increases the derivative at zero by $1+t$; the second preserves the entire first jet on a source disk larger by that factor. The common perturbation estimate of Proposition 4.10 gives linear control of both Area and B^2 . In particular,

$$|\text{Area}(G_t) - \text{Area}(f)| \leq C_{II}t, \quad |B(G_t)^2 - B(f)^2| \leq C_{II}t/\pi,$$

with constants uniform under the declared area and derivative caps. These inequalities give the cost of each enlargement, rather than only continuity for an individual disk. An alternative scalar repair has a linear cost in the absolute radius increment.

Under a fixed derivative cap, the two operations have common positive steps. A bounded loop alternates enlargement below the area window and descent above it, checking B after either surgery. It reaches every specified derivative threshold with the explicit counters of Proposition 2.9. The radii of the recorded disks may still have a finite limit. In that case marked compactness either gives a sphere meeting and escaping F , or retains a normalized principal map. A simultaneous point selection shows that the latter has lost at least one sphere-area quantum $e_X > 0$, including when concentration occurs at the source boundary.

Enlargement may increase area, so the loss at recovery cannot alone bound the total number of discarded bubbles. We instead record the recovered principal maps before enlarging them. A limit of these recorded maps lowers their common area cap by a further quantum. The repair costs are accounted for separately, and their radius increments force progress when the returned derivatives stay bounded. This yields a finite hierarchy of recoveries, reaching every finite target radius or producing the required sphere. Section 2.3 gives the construction and its depth bound.

Section 6 organizes recovery in another way. A finite cover selects a collection of replacement disks. Once this catalogue is chosen, explicit rescaling restores the derivative cap and lowers area. Together with the two surgeries, it gives a bounded iteration for each prescribed scalar-derivative target.

For expanding sources, relative compactness gives either the escaping sphere or a normalized finite-area entire principal map. Removal at infinity closes the latter to a sphere. A final two-marked limit retains a rational chain from F to the boundary of a fixed neighborhood while the area cap decreases to A_* . Its first component leaving F has the required incidence and area. We derive the needed fixed-coordinate and marked compactness consequences from Ivashkovich and Shevchishin [30] in Section 5.

Figure 2 depicts the passage from the expanded disks to the sphere and the final marked limit, for a point fiber.

Plan of the paper.

Section 2 proves the relative sphere theorem and records the metric sharpness example. The Ricci-trace calculation, uniform holomorphic realization, and marked compactness are established in Sections 3–5. Section 6 gives the finite-catalogue alternative. Section 7 applies classical algebraic results to deduce rational connectedness and Hartshorne’s conjecture over $\mathbb{C}$. The final section explains a shorter existence argument, the geometric reason for retaining explicit surgeries, and the relation to Mori’s method and numerical positivity.

2. EXPLICIT CONSTRUCTION OF A RELATIVE SPHERE

We construct the disks themselves and retain their area and weighted derivative costs. Finite stages reach any chosen derivative threshold. At a finite limiting radius we recover the principal map, record an area loss, and repair it before continuing. A finite bound on the depth of such recoveries gives the relative sphere.

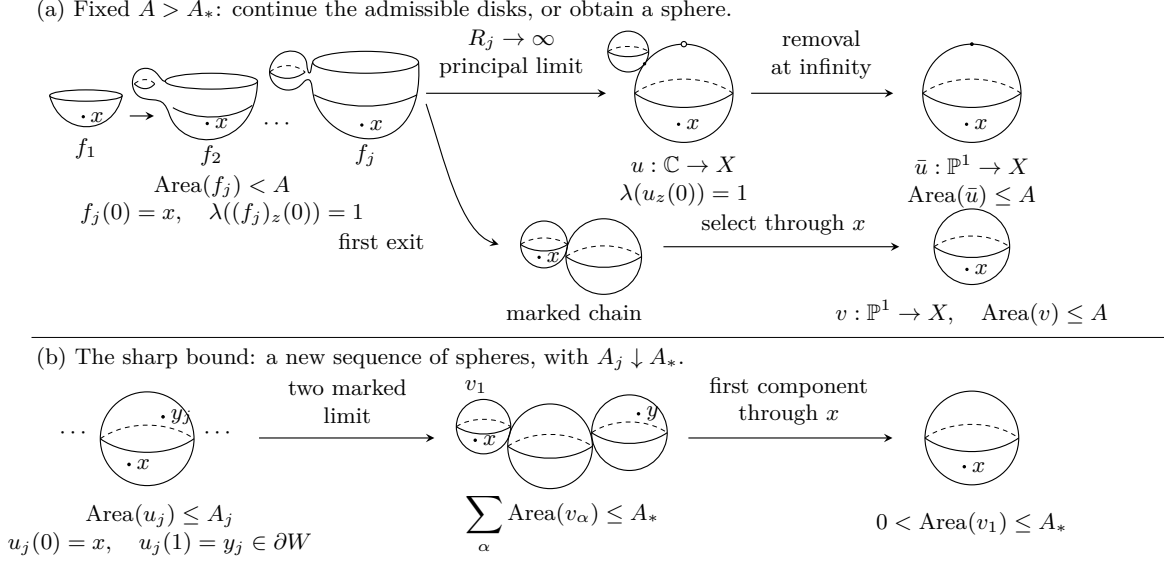


FIGURE 2. The two limiting stages, pictured for $F = \{x\}$. (a) Under a fixed cap, the compactness alternative supplies a marked chain and a sphere through x , or a normalized principal map. A normalized principal map is enlarged, restricted, and used to continue the construction; on expanding sources, finite-area removal closes the entire principal map at infinity. The bulge and neck illustrate possible concentration; a sphere at the node represents a bubble obtained after reparametrization. (b) Two marked values retain a chain from x to $y \in \partial W$ as $A_j \downarrow A_*$, where W is a relatively compact neighborhood of x with smooth boundary inside a coordinate chart. Its first component has area at most A_* ; it need not contain y or retain the derivative normalization. All contours are schematic.

2.1. Normalized disks and the two operations

We fix the metric normalization and recall the area and weighted derivative quantities from the Introduction.

Unless otherwise specified, (X, ω) is a compact Kähler manifold of complex dimension $n \geq 1$. In holomorphic coordinates $w^1, \dots, w^n$, let $h = (h_{\alpha\bar{\beta}})$ be the Gram matrix of the coordinate frame. Our Hermitian inner products are conjugate-linear in the first variable and complex-linear in the second, so $h_{\alpha\bar{\beta}} = \langle \partial/\partial w^\alpha, \partial/\partial w^\beta \rangle_\omega$ and a coefficient column c has squared norm $c^* h c$. This convention accounts for the transposed indices in the formula for ω below. For a holomorphic source map f , write $f_z = df(\partial/\partial z)$. With repeated coordinate indices summed and dA denoting Euclidean area measure, our normalizations are

$$\omega = \frac{i}{2} h_{\beta\bar{\alpha}} dw^\alpha \wedge d\bar{w}^\beta, \quad f^* \omega = |f_z|_h^2 dA, \quad \text{Ric}(\omega) = -i\partial\bar{\partial} \log \det h. \quad (1)$$

For a holomorphic map $f: S \rightarrow X$ from a Riemann surface, its ω -area is $\text{Area}(f) := \int_S f^* \omega$, the geometric area counted with multiplicity. For $a \in \mathbb{C}$ and $r > 0$, write $\mathbb{D}_r(a) := \{z \in \mathbb{C} : |z - a| < r\}$, and abbreviate $\mathbb{D}_R := \mathbb{D}_R(0)$ and $\mathbb{D} := \mathbb{D}_1(0)$; bars denote closure, and $\partial\mathbb{D}_R$ is the boundary circle. We use the area on $\mathbb{D}_R$ and the boundary-distance-weighted derivative bound

$$\text{Area}_R(f) := \int_{\mathbb{D}_R} f^* \omega, \quad B_R(f) := \sup_{z \in \mathbb{D}_R} (R - |z|) |f_z(z)|_\omega. \quad (2)$$

Thus $B_R(f)$ weights the derivative by its distance to the source boundary. Write $\mathcal{O}(V, X)$ for the holomorphic maps from an open set $V \subset \mathbb{C}$ to X , and $K \Subset U$ when the closure of K is compact

and contained in U . A holomorphic sphere is a map $\mathbb{P}^1 \rightarrow X$; a rational curve is the reduced image of a nonconstant such map. Smooth convergence means uniform convergence of all derivatives on compact source subsets.

Under the Ricci bound $\text{Ric}(\omega) \geq \kappa\omega$, $\kappa > 0$, set

$$A_* = \frac{2\pi(n+1)}{\kappa}. \quad (3)$$

Positive Ricci curvature makes the anticanonical bundle positive, and Kodaira's theorem makes X projective; see Remark 3.2. Fix a projective embedding for the two disk operations below.

For the relative constraint, start with $q : N \rightarrow Y$, y_0 , and the full compact fiber $F = q^{-1}(y_0)$ in Theorem 1.1. We first reduce to a proper submersion near this fiber. Choose $F \Subset U \Subset N$ with smooth boundary, keeping $\bar{U}$ inside the submersion neighborhood. The full-fiber hypothesis gives $y_0 \notin q(\partial U)$. We can therefore choose a coordinate ball Y' about y_0 such that $\bar{Y}' \cap q(\partial U) = \emptyset$ and $Y' \subset q(U)$. After replacing N by $U \cap q^{-1}(Y')$ and Y by Y' , the map q is proper: the inverse image of a compact subset of Y' is closed in $\bar{U}$ and avoids ∂U . In the restricted coordinate ball write $d_Y = \dim Y$ and let $\psi : N \rightarrow \mathbb{C}^{d_Y}$ be the coordinate expression of q , translated so that $\psi^{-1}(0) = F$. Fix $0 \neq \eta \in (\mathbb{C}^{d_Y})^*$ and set

$$\lambda^F = (\lambda_x^F)_{x \in F}, \quad \lambda_x^F := \eta \circ d\psi_x \in T_x^*X.$$

The center $f(0)$ may vary in F , but the base covector η remains fixed. Near zero, the condition $\lambda_{f(0)}^F(f_z(0)) = 1$ is precisely $(\eta \circ \psi \circ f)_z(0) = 1$. In particular, $d\psi_{f(0)}(f_z(0)) \neq 0$, so the initial tangent vector is not tangent to F . The submersion property makes λ_x^F nonzero for every $x \in F$. The classes below consist of maps on $\mathbb{D}_R$ that extend holomorphically to some open neighborhood of $\bar{\mathbb{D}}_R$. The neighborhood may depend on the map. Set

$$\mathcal{A}_R^{F, \lambda^F} = \left\{ \begin{array}{l} f \in \mathcal{O}(\mathbb{D}_R, X) : f \text{ extends holomorphically near } \bar{\mathbb{D}}_R, \\ f(0) \in F, \quad \lambda_{f(0)}^F(f_z(0)) = 1 \end{array} \right\}. \quad (4)$$

For a map defined only on the open disk, we use *normalized* for the same center and scalar conditions, without the boundary-extension requirement. Within each iteration, A and B denote the area and weighted derivative of the current map on its stated source disk.

Choose a local coordinate seed in this class and restrict its radius until its area is less than A_* . Put

$$\ell_F = \max_{x \in F} \|\lambda_x^F\|_{\omega, *}, \quad B_R(f) \geq R|f_z(0)|_\omega \geq R/\ell_F. \quad (5)$$

Surgeries I and II preserve the current center. Principal recovery and catalogue replacement may move it within F while preserving the base scalar. The maps are not required to extend one another. Fix also $F \Subset V \Subset N$ with smooth boundary.

Fix $c > 0$ with $\omega \geq c\iota^*\omega_{\text{FS}}$, normalizing a projective line to have Fubini–Study area π , and set

$$e_X = c\pi > 0. \quad (6)$$

Every nonconstant holomorphic sphere has area at least e_X , because its projective degree is a positive integer.

The first operation lowers area above the Ricci threshold. Its parameter and weighted derivative cost are uniform under the two declared caps. The second operation has a linear area cost and enlarges the source.

Proposition 2.1. *Assume $\text{Ric}(\omega) \geq \kappa\omega$ with $\kappa > 0$, and keep the fixed projective embedding. For every $A_0, B_0, \delta > 0$ there are $\sigma > 0$ and $\beta_I < \infty$ such that each $f \in \mathcal{A}_R^{F, \lambda^F}$ with*

$$A_* + \delta \leq \text{Area}_R(f) \leq A_0, \quad B_R(f) \leq B_0$$

admits a map $g \in \mathcal{A}_R^{F, \lambda^F}$ satisfying

$$g(0) = f(0), \quad \text{Area}_R(g) \leq \text{Area}_R(f) - \sigma, \quad |B_R(g)^2 - B_R(f)^2| \leq \beta_I. \quad (7)$$

The map is a member of the explicit affine-lift family in Theorem 4.8, selected by a real area Hessian and the sign of its first derivative. Formula (64) specifies its step and the guaranteed area decrease. The same constants work when the scalar derivative 1 is replaced by any prescribed value k , with that value preserved by the operation.

Lemma 2.2. Fix caps A_0, B_0 and the projective embedding. There are constants $0 < \bar{t} \leq 1$ and $C_{II} < \infty$, independent of the radius and the individual map, with the following property. Let $h : \mathbb{D}_R \rightarrow X$ be holomorphic on a finite open disk, with $\text{Area}_R(h) \leq A_0$ and $B_R(h) \leq B_0$. A holomorphic affine lift H on $\mathbb{D}_R$ gives the maps

$$h_t(z) = \Pi(H(z) + tzH'(0)), \quad G_t(z) = h_t(z/(1+t)), \quad 0 < t \leq \bar{t}. \quad (8)$$

The map G_t is defined on $\mathbb{D}_{(1+t)R}$, preserves the full first jet of h , and satisfies

$$|\text{Area}_{(1+t)R}(G_t) - \text{Area}_R(h)| \leq C_{II}t, \quad (9)$$

$$|B_{(1+t)R}(G_t)^2 - B_R(h)^2| \leq C_{II}t/\pi. \quad (10)$$

These same estimates compare h_t and h on $\mathbb{D}_R$, and $(h_t)_z(0) = (1+t)h_z(0)$. If h extends past its closed disk, the maps h_t, G_t extend past their corresponding closed disks. For an open-disk map, the restriction

$$g = G_t|_{\mathbb{D}_{R'}}, \quad R' = (1+t/2)R, \quad (11)$$

extends past its closed disk and satisfies

$$\text{Area}_{R'}(g) \leq \text{Area}_R(h) + C_{II}t, \quad B_{R'}(g)^2 \leq B_R(h)^2 + C_{II}t/\pi. \quad (12)$$

In particular, relative normalization is preserved by G_t and g .

Proposition 2.1 and Lemma 2.2 are proved in Section 4.4. The proofs give the parameter choices, the common B^2 costs, the arbitrary-scalar descent, and the open-disk lift.

Every finite-area map on a finite disk has finite B , by Lemma 2.7. Thus the second operation applies to each principal map recovered below, using its actual value of B . There is no asserted common bound for those values before recovery. The two-sided estimate (10) refers to the full enlarged source; after strict source restriction only the upper bound (12) is used.

Figure 3 separates the affine perturbation from the source dilation.

2.2. Principal limits and marked values

Compactness serves two purposes here: it produces a principal limit in the original source coordinate, and it preserves a connection between two marked target sets when bubbling occurs. Finite-area removal then turns an entire principal map into a holomorphic sphere. The principal, removal and two-marked statements require only a compact Kähler target. Their individual proof locations are specified below.

Proposition 2.3. Let (X, ω) be compact Kähler. For each j , let $R_j > 0$ and let $f_j : U_j \rightarrow X$ be holomorphic, where $U_j \subset \mathbb{C}$ is open and $\overline{\mathbb{D}_{R_j}} \Subset U_j$. Assume that $\sup_j \text{Area}_{R_j}(f_j) < \infty$, and set $A_\infty = \liminf_j \text{Area}_{R_j}(f_j)$. Choose $R_\infty \in (0, \infty]$ with $R_\infty \leq \liminf_j R_j$, interpreting $\mathbb{D}_\infty$ as $\mathbb{C}$. After passing to a subsequence realizing A_∞ and then to a further subsequence, there are a finite set $\Sigma_\infty \subset \mathbb{D}_{R_\infty}$ and a holomorphic map $f_\infty : \mathbb{D}_{R_\infty} \rightarrow X$ such that, in the original source coordinate,

$$f_j \longrightarrow f_\infty \quad \text{smoothly on compact subsets of } \mathbb{D}_{R_\infty} \setminus \Sigma_\infty, \quad \text{Area}_{R_\infty}(f_\infty) \leq A_\infty.$$

The principal limit f_∞ is holomorphic on all of $\mathbb{D}_{R_\infty}$ and may be constant. The asserted convergence of f_j holds only away from Σ_∞ ; convergence at its points is not claimed.

The proof of Proposition 2.3 is in Section 5.2.

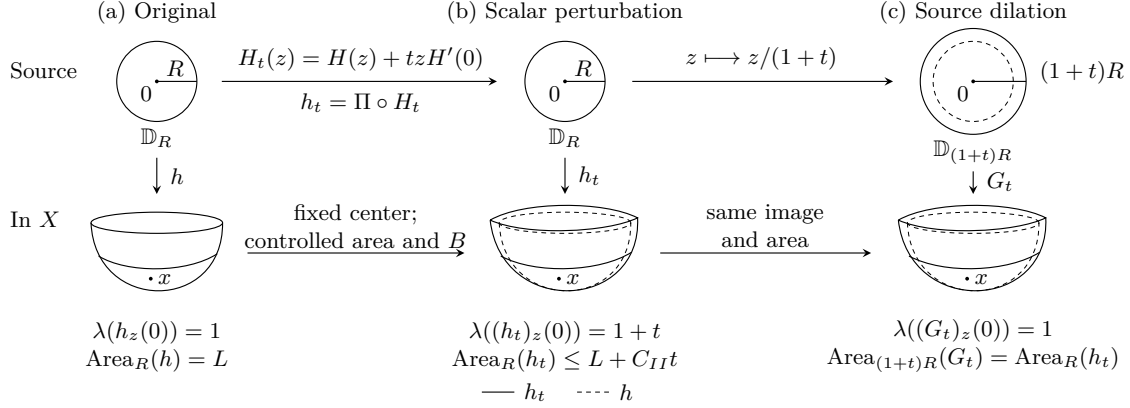


FIGURE 3. Enlargement of one finite-area disk at a fixed center $x \in F$, with $\lambda = \lambda_x^F$. The affine-lift perturbation $h_t = \Pi(H + tzH'(0))$ acts on the same disk and increases the scalar derivative from 1 to $1 + t$. Dilation gives $G_t(z) = h_t(z/(1+t))$ on $\mathbb{D}_{(1+t)R}$, with scalar derivative 1 and unchanged total area. The area and B^2 changes obey the common linear bounds in Lemma 2.2. The outlines are schematic; dashed target outlines show h .

Theorem 2.4. *Let X be compact Kähler. A holomorphic map from a punctured disk to X with finite area extends holomorphically over the puncture. Consequently, a finite-area holomorphic map $u : \mathbb{C} \rightarrow X$ extends to a holomorphic map $\mathbb{P}^1 \rightarrow X$.*

The proof of Theorem 2.4 is in Section 5.1.

Lemma 2.5. *Let K_0, K_1 be disjoint compact subsets of a compact Kähler manifold X . Suppose $R_j > 1$, $R_j \rightarrow \infty$, and the holomorphic maps $f_j : U_j \rightarrow X$, where $U_j \subset \mathbb{C}$ is open and $\overline{\mathbb{D}}_{R_j} \Subset U_j$, satisfy*

$$f_j(0) \in K_0, \quad f_j(1) \in K_1, \quad \sup_j \text{Area}_{R_j}(f_j) < \infty.$$

Then there are nonconstant holomorphic maps $u_1, \dots, u_\ell : \mathbb{P}^1 \rightarrow X$ whose images form a chain: the first meets K_0 , the last meets K_1 , and consecutive images intersect. Their total area, counted with the multiplicities of these maps, satisfies

$$\sum_{i=1}^{\ell} \text{Area}(u_i) \leq \liminf_j \text{Area}_{R_j}(f_j).$$

The proof of Lemma 2.5 is in Section 5.3.

The fixed source marks and the disjoint target sets play different roles. Since the marks remain at 0 and 1, stabilization cannot produce an attached tree made entirely of constant components: such a stable tree would need at least two marks converging to the same source point. Since K_0 and K_1 are disjoint, the nodal path between the marked values must contain a nonconstant component.

The scalar derivative is a derivative of the bounded local base map. This gives a direct alternative: a first exit from a neighborhood of F produces a sphere, while trapping near F allows the scalar constraint to pass to the principal map even if the original maps bubble at zero.

Lemma 2.6. *Let (X, ω) be compact Kähler and retain F , N , ψ , and λ^F from (4). Suppose $R_j \rightarrow R_\infty \in (0, \infty]$ and $f_j \in \mathcal{A}_{R_j}^{F, \lambda^F}$, with*

$$\text{Area}_{R_j}(f_j) \rightarrow E < \infty.$$

Then at least one of the following conclusions holds:

(i) *There is a nonconstant sphere $v : \mathbb{P}^1 \rightarrow X$ such that*

$$v(\mathbb{P}^1) \cap F \neq \emptyset, \quad v(\mathbb{P}^1) \not\subset F, \quad \text{Area}(v) \leq E.$$

(ii) *A subsequence has a principal limit $h : \mathbb{D}_{R_\infty} \rightarrow X$ in the original source coordinate, as in Proposition 2.3, satisfying*

$$\text{Area}_{R_\infty}(h) \leq E, \quad h(0) \in F, \quad \lambda_{h(0)}^F(h_z(0)) = 1.$$

On some fixed disk about zero, the base compositions $\psi \circ f_j$ converge to $\psi \circ h$ smoothly on compact subsets. Here $\mathbb{D}_\infty = \mathbb{C}$.

Proof. Choose $F \Subset V \Subset N$ with smooth boundary, and fix $r > 0$ such that $R_j > r$ for all sufficiently large j . There are two possibilities after passage to a subsequence.

First, suppose there are $z_j \rightarrow 0$ with $z_j \neq 0$ and $f_j(z_j) \in \partial V$. Such points can be chosen as first exits along segments from zero whenever no fixed small closed source disk has image in V along an infinite subsequence. The rescaled maps

$$v_j(\zeta) := f_j(z_j \zeta), \quad |\zeta| < R_j/|z_j|,$$

extend past their closed source disks and satisfy

$$v_j(0) \in F, \quad v_j(1) \in \partial V, \quad \text{Area}_{R_j/|z_j|}(v_j) = \text{Area}_{R_j}(f_j) \rightarrow E.$$

Their source radii tend to infinity. Lemma 2.5 gives a chain of nonconstant spheres from F to ∂V , of total area at most E . Take its first component not contained in F . It meets F , either at the first marked value or at its intersection with the preceding component, and has area at most E . This proves (i).

Otherwise, for some $0 < \rho < r$, an infinite subsequence satisfies $f_j(\overline{\mathbb{D}_\rho}) \subset V$. Apply Proposition 2.3 on $\mathbb{D}_{R_\infty}$ to this subsequence. Its principal map h takes values in $\overline{V}$ on $\mathbb{D}_\rho$: this holds away from the finite concentration set by convergence, and at that set by continuity. In particular, $\psi \circ h$ is defined there. The bounded holomorphic maps $\psi \circ f_j$ have a further locally uniformly convergent subsequence. Principal convergence identifies its limit with $\psi \circ h$ away from the concentration set; the identity theorem identifies them throughout $\mathbb{D}_\rho$. Cauchy estimates give smooth convergence on compact subdisks. Passing the value and scalar derivative at zero to the limit yields

$$\psi(h(0)) = 0, \quad (\eta \circ \psi \circ h)_z(0) = \lim_j (\eta \circ \psi \circ f_j)_z(0) = 1.$$

The area bound follows from principal compactness, proving (ii). $\square$

The trapped branch also retains a quantified amount of scalar base area. On its fixed disk $\mathbb{D}_\rho$, set $b_j = \eta \circ \psi \circ f_j$ and $b = \eta \circ \psi \circ h$. For every $0 < r < \rho$, the submean inequality and $b'_j(0) = 1$ give

$$\int_{\mathbb{D}_r} |b'_j|^2 dA \geq \pi r^2, \quad \int_{\mathbb{D}_r} |b'|^2 dA = \lim_j \int_{\mathbb{D}_r} |b'_j|^2 dA \geq \pi r^2. \quad (13)$$

The equality follows from smooth convergence on the smaller disk. This is area in the fixed scalar base coordinate. It is carried by the retained principal map and does not locate a bubble selected elsewhere.

The following lemma supplies the weighted derivative bound for the principal map at a finite endpoint.

Lemma 2.7. *Let (X, ω) be compact Kähler, let $0 < R < \infty$, and let $h : \mathbb{D}_R \rightarrow X$ be holomorphic with $\text{Area}_R(h) < \infty$. Then $B_R(h) < \infty$.*

The proof in Section 5.4 uses point selection: an unbounded weighted derivative for this fixed map would place a positive amount of area in arbitrarily thin boundary collars.

Recovery with an area loss.

Lemma 2.8. *Let f_j be normalized maps holomorphic near $\overline{\mathbb{D}}_{R_j}$, where $R_j \rightarrow R \in (0, \infty)$, $\text{Area}_{R_j}(f_j) \rightarrow E < \infty$, and $B_{R_j}(f_j) \rightarrow \infty$. One may select either a sphere meeting and escaping F of area at most E , or a normalized principal map $h : \mathbb{D}_R \rightarrow X$ satisfying*

$$\text{Area}_R(h) \leq E - e_X, \quad B_R(h) < \infty. \quad (14)$$

The assertion also holds for finite-area maps on the open disks, without extension across the boundary.

Proof. Apply the relative compactness alternative just stated. Retain the principal branch when that is the selected alternative, and its subsequence. Use the point selection of Lemma 5.2 on this same subsequence, obtaining a nonconstant sphere u . If w_j is its selected point and $\Lambda_j = |(f_j)_z(w_j)|_\omega$, then

$$\Lambda_j \geq B_{R_j}(f_j)/R_j \rightarrow \infty.$$

Pass to a subsequence with $w_j \rightarrow w \in \overline{\mathbb{D}}_R$. The point w lies in the finite concentration set of the principal convergence or on $\partial\mathbb{D}_R$: local smooth convergence at any other interior point bounds the derivatives there.

Let Q be a compact source region avoiding that concentration set, and fix $S < \infty$. For all sufficiently large j , Q is disjoint from $\mathbb{D}_{S/\Lambda_j}(w_j)$, the source region representing $u_j|_{\mathbb{D}_S}$. Both regions lie in $\mathbb{D}_{R_j}$. Their areas therefore give, on taking the limit,

$$\int_Q h^* \omega + \int_{\mathbb{D}_S} u^* \omega \leq E.$$

First exhaust the principal disk outside its finite exceptional set and then let $S \rightarrow \infty$. The result is $\text{Area}_R(h) + \text{Area}(u) \leq E$, including when w is a source boundary point. Equation (6) proves the first inequality in (14). A finite-area holomorphic map on a finite disk has finite weighted derivative by Lemma 2.7.

For open-disk maps choose a point w_j with $(R_j - |w_j|)|(f_j)_z(w_j)| \geq B_{R_j}(f_j)/2$, and restrict to a radius $r_j < R_j$ so close to R_j that

$$R_j - r_j < 1/j, \quad \text{Area}_{R_j}(f_j) - \text{Area}_{r_j}(f_j) < 1/j, \quad r_j - |w_j| \geq (R_j - |w_j|)/2.$$

The restricted maps extend past their closed disks, have radii tending to R , areas tending to E , and weighted derivatives tending to infinity. The preceding proof applies to these maps. $\square$

This recovery retains a whole principal map, discards the rescaled sphere, and permits the center to move in F . It requires a sequence and a subsequential limit. It is not a cutting and gluing operation on one given disk, and it makes no claim that the recovered map is close to that disk near a bubble.

2.3. Iteration and the relative sphere

Fix $\delta > 0$ and set

$$L = A_* + \delta, \quad \eta_0 = \delta/2, \quad C = A_* + 2\delta. \quad (15)$$

Keep the normalized coordinate seed already chosen above.

Fix a stage threshold $M > 0$. All the constants in this stage are chosen using the same caps C, M . Use σ from (64) and, for the affine enlargement, choose once and for all

$$\tau_M = \min \left\{ \frac{\bar{t}}{2}, \frac{\eta_0}{\max\{1, C_{II}\}} \right\} > 0. \quad (16)$$

One enlargement then multiplies the radius by $1 + \tau_M$ and increases area by at most η_0 . Given the current disk, follow these rules:

- (1) If $B_R(f) \geq M$, record this disk and end the stage.
- (2) If $B_R(f) < M$ and $\text{Area}_R(f) \geq L$, perform Surgery I and return to the first rule.

- (3) If $B_R(f) < M$ and $\text{Area}_R(f) < L$, perform Surgery II with $t = \tau_M$ and return to the first rule.

Every comparison is made with the actual current map. In particular a descent step which increases B past M ends the stage immediately. It is neither discarded nor subjected to an operation requiring $B < M$.

Proposition 2.9. *Let the starting disk of a stage have radius R_s , area $A_s < C$, and satisfy (4). The preceding rules produce a disk with $A < C$ and $B \geq M$ in finitely many surgeries. If $B_{R_s}(f_s) < M$, define*

$$N_M = \left\lceil \frac{\log(\ell_F M / R_s)}{\log(1 + \tau_M)} \right\rceil, \quad (17)$$

$$J_0 = \begin{cases} 0, & A_s < L, \\ \lfloor (A_s - L)/\sigma \rfloor + 1, & A_s \geq L, \end{cases} \quad (18)$$

$$J = \lfloor \eta_0 / \sigma \rfloor + 1. \quad (19)$$

An explicit upper bound for the total number of surgeries is

$$T_M = J_0 + N_M(1 + J). \quad (20)$$

If the starting disk already has $B \geq M$, no surgery is required.

Proof. The value and scalar derivative are preserved by both operations. Descent lowers area. Enlargement is used only below L , so its new area is less than $L + \eta_0 < C$. This proves the common area bound at every step. Before any operation the first rule has checked $B < M$, so the same constants chosen at the beginning of the stage apply.

We may implement the rules using bounded loops. First allow at most J_0 descents, stopping immediately upon reaching $B \geq M$ or $A < L$. If all J_0 descents are used, the latter condition follows from their guaranteed decrease; it does not require an infinite iteration argument. Thereafter each enlargement starts below L and ends below $L + \eta_0$. At most J subsequent descents suffice to restore $A < L$, unless the derivative threshold has already been met.

Allow at most N_M such enlargement-and-descent blocks. At the N_M th enlargement, if this block is needed, the radius is

$$R_s(1 + \tau_M)^{N_M} \geq \ell_F M.$$

Equation (5) then certifies $B \geq M$, and the stage stops. Thus the stated finite loops always furnish the required disk. Equality in either threshold is covered by the tests $B \geq M$ and $A \geq L$, and the extra 1 in each descent count handles equality at the area threshold.

There is also a useful cumulative area account. If N_I, N_{II} denote the two operation counts along any completed prefix, then

$$0 \leq A_{\text{current}} \leq A_s - N_I \sigma + N_{II} \eta_0, \quad N_I \sigma \leq A_s + N_{II} \eta_0. \quad (21)$$

Thus the area introduced by enlargements is explicitly charged in the bound for the number of descents. $\square$

This is the distinction between a possible next step and uniform progress: each fixed stage has a common positive enlargement, a common positive area decrease, and a finite bound on the total number of operations. Neither the derivative quantity nor area is required to be monotone throughout the entire recursion.

A finite hierarchy of principal recoveries.

The energy loss in Lemma 2.8 does not bound the total number of recoveries in a history that also uses Surgery II: the enlargements can pay for new bubbles. We instead count the *depth of successive limits of recovered maps*. This gives a finite hierarchy of countable procedures. A selection of a convergent subsequence is an allowed analytic operation throughout this section. At each level we

record principal maps before spending area on their enlargement, and take limits of those recorded maps.

Keep C, L, η_0 from (15). Fix a finite target radius $T > 0$. The initial map of every procedure is normalized, extends past its closed disk, and has area strictly less than C . The permitted results are:

- (1) a sphere meeting and escaping F , of area at most C ;
- (2) a normalized disk with source radius at least T and area at most C ; this terminal disk may be defined only on an open disk;
- (3) a normalized finite-area principal map on a radius $R < T$, with a strictly specified lower area cap. This third result is passed to the next level of the construction.

Radii never decrease during the procedures or the repairs between them.

Proposition 2.10. *For each integer $m \geq 0$ there is a procedure $\mathcal{P}_m$ with the preceding initial data and alternatives, whose third result, if selected, has*

$$\text{Area}_R(h) \leq C - (m + 1)e_X. \quad (22)$$

It uses the explicit finite stages of Section 2.3, the repair (11), and countable sequences with the recovery of Lemma 2.8. There are at most $m + 1$ nested levels of this recovery construction.

Proof.

Level zero.

If the initial radius is already at least T , retain the initial map. Otherwise run the finite threshold procedure with $M = 1, 2, 3, \dots$, carrying the current closed disk to the next stage. After each surgery, and at each recorded stage, stop if the radius is at least T . If that happens, the second result has been obtained. The finite stages are valid in the relative normalization as explained above. Every map has area less than C .

If all stages are performed, their recorded maps satisfy $B_{R_j}(f_j) \geq j$. Their nondecreasing radii have a limit $R \in (0, T]$. Choose a subsequence whose areas tend to $E \leq C$. Lemma 2.8 provides either the first result or a normalized principal map with area at most $C - e_X$. If $R = T$, retain this principal map as the second result. If $R < T$, retain it as the third result. This defines $\mathcal{P}_0$.

One additional level.

Suppose $\mathcal{P}_{m-1}$ has been defined, with $m \geq 1$. Apply it to the current closed disk. Its first or second result is also the corresponding result of $\mathcal{P}_m$. For a third result write (h_j, R_j) for the returned principal map and its radius. By the induction hypothesis,

$$R_j < T, \quad \text{Area}_{R_j}(h_j) \leq C - me_X. \quad (23)$$

The derivative quantity $b_j = B_{R_j}(h_j)$ is finite.

Preselect the constants of Lemma 2.2 for each integer derivative cap $q \geq 1$ and the one area cap C . Denote them by $\bar{t}_q, C_{II,q}$, and put

$$q_j = \max\{1, \lceil b_j \rceil\}, \quad t(q) = \min \left\{ \bar{t}_q/2, \frac{e_X}{2 \max\{1, C_{II,q}\}} \right\} > 0. \quad (24)$$

Apply the affine repair with $t = t(q_j)$ and retain the closed disk of radius

$$R'_j = (1 + t(q_j)/2)R_j.$$

Its area is at most $C - me_X + e_X/2 < C$, and its new B satisfies (12). Thus it is an admissible initial map for another call to $\mathcal{P}_{m-1}$. If $R'_j \geq T$, stop with the second result. Otherwise make that call and repeat this rule. Every lower-level call starts with a closed disk; no open-disk version of Surgery I is being assumed.

If a lower-level call terminates with the first or second result, retain it. The remaining case is a countable sequence of principal returns (h_j, R_j) satisfying (23). The intervening repairs imply

$$R_{j+1} \geq (1 + t(q_j)/2)R_j. \quad (25)$$

These radii increase to $R \leq T$.

Here there is an explicit way to select large- B returns. For any integer $Q \geq 1$, set

$$a_Q = \min_{1 \leq q \leq Q} t(q)/2 > 0.$$

If $b_j \leq Q$ at each of s consecutive returns, their radii increase by a factor at least $(1 + a_Q)^s$. Starting at any such return, after at most

$$1 + \left\lceil \frac{\log(T/R_j)}{\log(1 + a_Q)} \right\rceil$$

further returns the procedure has either terminated or recorded $b > Q$. Applied successively with $Q = 1, 2, 3, \dots$, starting each scan strictly after the previously selected index, this rule selects a subsequence of returns with $b_j \rightarrow \infty$ whenever the countable sequence persists. Its radii still tend to R .

Select a further subsequence with areas tending to E . Crucially, these are the *returned principal maps*, before their subsequent enlargements. Equation (23) gives $E \leq C - me_X$. The open-disk part of Lemma 2.8 provides either a sphere as in the first result or a normalized principal map with

$$\text{Area}_R(h) \leq E - e_X \leq C - (m + 1)e_X.$$

Retain it as the second result if $R = T$, and as the third result if $R < T$. This completes the recursive definition of $\mathcal{P}_m$. It adds one countable sequence and one recovery level to the already defined procedure $\mathcal{P}_{m-1}$. $\square$

Corollary 2.11. *Put $N_C = \lfloor C/e_X \rfloor$. For every normalized initial closed disk of area less than C and every $T < \infty$, the procedure $\mathcal{P}_{N_C}$ gives either a sphere meeting and escaping F of area at most C , or a normalized disk of radius at least T and area at most C .*

Proof. The third result at this specified level would have area at most $C - (N_C + 1)e_X < 0$, so it is not among the possible returned maps. The first two alternatives are precisely the asserted results. $\square$

The finite bound concerns the *number of nested recovery levels*, not the number of bubbles discarded over the whole history. Each level may make countably many calls to the preceding level. Since the depth is finite, the construction uses only countably many individual surgeries and selections of subsequences. No uncountable recursion, maximal-radius profile, or choice of almost minimizers is used. The proof never assumes that escaping rational curves are absent: whenever the compactness operation supplies one, the procedure stops and retains it. The last corollary is a finite rank bound based on nonnegativity of area.

The ledger that makes the construction work is

$$\underbrace{\text{Area}(h_j) \leq C - me_X}_{\text{returned maps at one level}} \quad \longrightarrow \quad \underbrace{\text{Area}(h) \leq C - (m + 1)e_X}_{\text{limit of those returned maps}}.$$

Area added between these returns does not enter this particular sequence of maps. It is bounded separately by $e_X/2$ for each repair and keeps the next closed disk strictly below C . This is why unbounded accumulated enlargement cost does not invalidate the depth bound.

Proof of Theorem 1.1. Keep the proper neighborhood, normalization and seed chosen above.

Fix $\delta > 0$ and $C = A_* + 2\delta$, and retain the neighborhood V already chosen. For $j = 1, 2, 3, \dots$, independently apply Corollary 2.11 to this seed with target $T = 2j$ and cap $C = A_* + 2\delta$. Stop whenever a sphere is returned. If disk results are returned for every j , restrict each one to $\mathbb{D}_j$.

These restrictions extend past their closed disks, are normalized, and have area at most C . Select a subsequence whose areas converge. Their source radii tend to infinity. The relative compactness alternative supplies either the desired sphere directly or a normalized entire principal map $u : \mathbb{C} \rightarrow X$ of area at most C . Finite-area removal extends the latter across infinity. Its scalar base derivative is 1 at zero, so it meets F and is not contained in F .

To obtain the exact bound, repeat the construction with $\delta_j = 1/(2j)$, obtaining spheres of areas at most $A_* + 1/j$. Each sphere meets both F and ∂V . Indeed, if its image were contained in V , the holomorphic map $\psi \circ u$ from $\mathbb{P}^1$ to $\mathbb{C}^{d_V}$ would be constant, and its value at a point of F would force the image to lie in F . Select a subsequence whose sphere areas converge; its limit is at most A_* . Choose one point mapped to each of F and ∂V , and reparametrize the sphere so these distinct source points are 0 and 1. Restrict to disks of radii tending to infinity. Lemma 2.5, with the disjoint compact sets F and ∂V , gives a rational chain between these sets of total area at most A_* . Its first component not contained in F meets F , at the first marked value or at the node joining it to the preceding component. Retain that component. $\square$

Proof of Theorem 1.2. Given $x \in X$, let N be a coordinate neighborhood of x and let $q : N \rightarrow Y \subset \mathbb{C}^n$ be its coordinate map. The full fiber over $q(x)$ is $F = \{x\}$, and q is a submersion. Theorem 1.1 therefore gives a nonconstant sphere through x with area at most A_* . $\square$

Remark 2.12. On $\mathbb{P}^n$, use the Fubini–Study form

$$\omega_{\text{FS}} = \frac{i}{2} \partial \bar{\partial} \log(1 + |w^1|^2 + \cdots + |w^n|^2).$$

In the normalization (1), $\text{Ric}(\omega_{\text{FS}}) = 2(n+1)\omega_{\text{FS}}$ and $\int_{\mathbb{P}^1} \omega_{\text{FS}} = \pi$ on a projective line. Every nonconstant sphere in $\mathbb{P}^n$ has positive integer degree d and area $d\pi$. Thus for $\kappa = 2(n+1)$ the bound $2\pi(n+1)/\kappa = \pi$ is attained by a line and cannot be decreased.

3. RICCI TRACES AND SCALAR-CONSTRAINED VARIATIONS

The Ricci trace separates curvature from the cost of fixing a value and one scalar derivative. An adapted multiplier has Dirichlet cost $\pi(n+1)$; Proposition 3.4 turns this into the descent estimate. Section 4 realizes the variation fields by holomorphic families. We retain the normalizations (1)–(2).

For $z = x + iy$, the complex derivative and the Euclidean Laplacian are

$$\partial_z = \frac{1}{2}(\partial_x - i\partial_y), \quad \Delta = \partial_x^2 + \partial_y^2 = 4\partial_z\partial_{\bar{z}}.$$

On f^*T_X , ∇_z denotes the Chern covariant derivative induced by ω in the source direction. On $\partial\mathbb{D}_R$, ν is the outward Euclidean unit normal, ∂_ν the corresponding derivative, and ds Euclidean arclength. Let A^* denote the conjugate transpose of A . We use the Hilbert–Schmidt norm and the operator norm

$$\|A\|_{\text{HS}}^2 := \text{tr}(AA^*) = \sum_{j,k} |A_{jk}|^2, \quad \|A\|_{\text{op}} := \sup_{|v|=1} |Av|. \quad (26)$$

The symbol Id_n denotes the identity matrix. Distances and diameters use the specified metric.

For a Hermitian holomorphic line bundle (L, h) , we use the real Chern curvature form $-i\partial\bar{\partial} \log |s|_h^2$, where s is a nonvanishing local holomorphic frame. Its cohomology class is $2\pi c_1(L)$. We write $c_1(X) = c_1(K_X^{-1})$.

Yau’s theorem supplies the Ricci lower bound that allows the relative sphere theorem to be applied to Fano manifolds.

Proposition 3.1. *Every complex Fano manifold admits a Kähler metric ω and a constant $\kappa > 0$ such that*

$$\text{Ric}(\omega) \geq \kappa\omega. \quad (27)$$

Proof. Ampleness of K_X^{-1} gives a positive closed $(1, 1)$ -form ρ representing $2\pi c_1(X)$. In a fixed Kähler class, Yau's prescribed-Ricci theorem [48] gives a metric ω with $\text{Ric}(\omega) = \rho$. The least eigenvalue of ρ relative to ω is positive and continuous on the compact manifold. Its positive minimum therefore supplies κ with $\text{Ric}(\omega) \geq \kappa\omega$. $\square$

Remark 3.2. A connected compact Kähler manifold with a metric satisfying (27) is Fano. The induced Hermitian metric on K_X^{-1} has positive Chern curvature $\text{Ric}(\omega)$, so Kodaira's embedding theorem makes K_X^{-1} ample and X projective [33, pp. 314–316].

For a holomorphic family on a fixed source disk, the Kähler condition expresses the paired second area variation as a boundary integral. The boundary values are free, and no critical-point assumption is made. This identity lets the trace calculation use variation fields before constructing their holomorphic families.

Lemma 3.3. *Fix $R > 0$. Let $\Delta_t \subset \mathbb{C}$ be a parameter disk centered at zero, $U \subset \mathbb{C}$ an open neighborhood of $\overline{\mathbb{D}}_R$, and $\mathcal{F} : U \times \Delta_t \rightarrow X$ a holomorphic map. Write $f_t(z) := \mathcal{F}(z, t)$ and $\xi := \partial_t \mathcal{F}(\cdot, 0)$. Denote the second derivatives along the two real parameter axes by*

$$H_{\mathbb{R}} := \left. \frac{d^2}{ds^2} \right|_{s=0} \text{Area}_R(f_s), \quad H_{i\mathbb{R}} := \left. \frac{d^2}{ds^2} \right|_{s=0} \text{Area}_R(f_{is}).$$

Then

$$H_{\mathbb{R}} + H_{i\mathbb{R}} = \int_{\partial \mathbb{D}_R} \partial_\nu |\xi|_\omega^2 ds. \quad (28)$$

For a fixed central map f_0 , the sum $H_{\mathbb{R}} + H_{i\mathbb{R}}$ therefore depends only on the variation field ξ , not on the second-order parameter data of the family.

Proof. Set $\tilde{\omega} := \mathcal{F}^* \omega$, and write its Hermitian coefficient matrix in the source coordinates $\zeta^1 = z$, $\zeta^2 = t$ as

$$\tilde{\omega} = \frac{i}{2} \sum_{\alpha, \beta=1}^2 \tilde{\omega}_{\alpha\bar{\beta}} d\zeta^\alpha \wedge d\bar{\zeta}^\beta.$$

Since $d\tilde{\omega} = 0$, comparison of its $(2, 1)$ and $(1, 2)$ coefficients gives

$$\partial_t \tilde{\omega}_{z\bar{z}} = \partial_z \tilde{\omega}_{t\bar{z}}, \quad \partial_{\bar{t}} \tilde{\omega}_{t\bar{z}} = \partial_z \tilde{\omega}_{t\bar{t}}.$$

Differentiating and substituting gives

$$\partial_t \partial_{\bar{t}} \tilde{\omega}_{z\bar{z}} = \partial_z \partial_{\bar{t}} \tilde{\omega}_{t\bar{z}} = \partial_z \partial_{\bar{z}} \tilde{\omega}_{t\bar{t}}.$$

At $t = 0$, $\tilde{\omega}_{t\bar{t}} = |\xi|_\omega^2$. Consequently

$$\partial_t \partial_{\bar{t}} \text{Area}_R(f_t)|_{t=0} = \int_{\mathbb{D}_R} \partial_z \partial_{\bar{z}} |\xi|_\omega^2 dA = \frac{1}{4} \int_{\partial \mathbb{D}_R} \partial_\nu |\xi|_\omega^2 ds.$$

The sum of the second derivatives along the real and imaginary parameter axes is the parameter Laplacian $4\partial_t \partial_{\bar{t}}$. Hence

$$H_{\mathbb{R}} + H_{i\mathbb{R}} = 4 \partial_t \partial_{\bar{t}} \text{Area}_R(f_t)|_{t=0},$$

which yields (28). $\square$

Fix $R > 0$ and a map f holomorphic on a neighborhood of $\overline{\mathbb{D}}_R$. Without choosing a deformation family, we can define the following boundary quadratic form on holomorphic sections ξ of $f^* T_X$ that are smooth on the closed disk:

$$\mathcal{H}_{R,f}(\xi) := \int_{\partial \mathbb{D}_R} \partial_\nu |\xi|_\omega^2 ds. \quad (29)$$

When ξ is the tangent field of a complex holomorphic family, Lemma 3.3 identifies $\mathcal{H}_{R,f}(\xi)$ with the sum of the second area derivatives along the real and imaginary parameter axes.

A boundary-unitary holomorphic frame, supplied by Lemma 4.1, makes the curvature term a Ricci integral after summation. The multiplier contributes only its Dirichlet cost, so a suitable interpolation multiplier will turn the identity into the descent threshold.

Proposition 3.4. *Let $e = (e_1, \dots, e_n)$ be a holomorphic frame of f^*T_X , smooth on the closed disk and unitary on $\partial\mathbb{D}_R$. For a positive integer N , let Φ be a holomorphic $\text{Mat}_{n \times N}(\mathbb{C})$ -valued map on a neighborhood of $\mathbb{D}_R$, satisfying the boundary coisometry condition*

$$\Phi(\zeta)\Phi(\zeta)^* = \text{Id}_n \quad (|\zeta| = R).$$

The columns Φ_a give sections $(e\Phi)_a = \sum_{j=1}^n e_j \Phi_{ja}$ satisfying the trace identity

$$\sum_{a=1}^N \mathcal{H}_{R,f}((e\Phi)_a) = 4 \int_{\mathbb{D}_R} \|\Phi'\|_{\text{HS}}^2 dA - 2 \int_{\mathbb{D}_R} f^* \text{Ric}(\omega). \quad (30)$$

Here Φ' is the entrywise derivative and $\|\cdot\|_{\text{HS}}$ is the norm in (26).

Proof. With our convention that the metric is linear in its second variable, the Gram matrix $G := (\langle e_j, e_k \rangle)_{j,k}$ represents squared norms by $|ec|_\omega^2 = c^* G c$. Summing (29) over the columns gives the boundary integral of $\partial_\nu \text{tr}(G\Phi\Phi^*)$. On the boundary, $G = \Phi\Phi^* = \text{Id}_n$, but their normal derivatives need not vanish. The product rule and Jacobi's determinant identity therefore give, on $\partial\mathbb{D}_R$,

$$\begin{aligned} \partial_\nu \text{tr}(G\Phi\Phi^*) &= \text{tr}(\partial_\nu G) + \text{tr}(\partial_\nu(\Phi\Phi^*)), \\ \partial_\nu \log \det G &= \text{tr}(G^{-1} \partial_\nu G) = \text{tr}(\partial_\nu G). \end{aligned}$$

The divergence theorem and $\Delta \text{tr}(\Phi\Phi^*) = 4\|\Phi'\|_{\text{HS}}^2$ give the multiplier term. Since $\det G$ is the metric coefficient on $f^*K_X^{-1}$, (1) gives

$$f^* \text{Ric}(\omega) = -\frac{1}{2} \Delta \log \det G dA.$$

Green's formula therefore turns the determinant term $\partial_\nu \log \det G$ into $-2 \int f^* \text{Ric}(\omega)$. $\square$

We first construct variations that fix the value of a disk at 0 and one nonzero scalar component of its derivative. Fix $x_0 \in X$ and a nonzero complex-linear covector

$$\lambda \in T_{x_0}^* X \setminus \{0\}. \quad (31)$$

The scalar condition $\lambda(f_z(0)) = 1$ fixes one component of the initial derivative and gives $|f_z(0)|_\omega \geq \|\lambda\|_{\omega,*}^{-1}$, where the norm on the covector is dual to ω at x_0 . The multiplier below imposes the linearized value and scalar-derivative constraints. Proposition 2.1 realizes these constraints by holomorphic families that preserve $f(0) = x_0$ and $\lambda(f_z(0)) = 1$ exactly, while leaving the other derivative components free. In Section 2, the center varies in a compact fiber and the covector is pulled back from the base. In the principal branch of Lemma 2.6, this base normalization passes to the limit. For a disk satisfying $f(0) = x_0$ and $\lambda(f_z(0)) = 1$, the frame value $e(0) : \mathbb{C}^n \rightarrow T_{x_0} X$ identifies the scalar constraint with the nonzero covector $L_f := \lambda \circ e(0)$. Choose $U_f \in U(n)$ so that its last $n-1$ columns form an orthonormal basis of $\ker L_f$. This leaves one direction subject to the scalar constraint. In this adapted basis, set

$$\Phi_f(z) := U_f \text{diag} \left((z/R)^2, z/R, \dots, z/R \right). \quad (32)$$

The columns vanish at zero, so $(e\Phi_f)'(0) = e(0)\Phi_f'(0)$. The first column has zero derivative; the others have derivatives in $\ker L_f$. Thus $\Phi_f(0) = 0$ and $L_f \Phi_f'(0) = 0$ impose the infinitesimal constraints. On $|z| = R$, the diagonal entries have modulus one, so $\Phi_f \Phi_f^* = \text{Id}_n$. For each integer $d \geq 1$, scaling $z = R\zeta$ and integrating in polar coordinates gives $\int_{\mathbb{D}_R} |((z/R)^d)'|^2 dA = \pi d$,

independently of R . Since U_f is unitary, the quadratic column contributes 2π and each of the $n - 1$ linear columns contributes π . Hence

$$\int_{\mathbb{D}_R} \|\Phi'_f\|_{\text{HS}}^2 dA = 2\pi + (n - 1)\pi = \pi(n + 1). \quad (33)$$

If $\text{Ric}(\omega) \geq \kappa\omega$, then

$$\sum_{a=1}^n \mathcal{H}_{R,f}((e\Phi_f)_a) \leq 4\pi(n + 1) - 2\kappa \text{Area}_R(f). \quad (34)$$

The cost in (33) is optimal within the prescribed multiplier class. Indeed, fix $R > 0$, $N \geq 1$, and $0 \neq L \in (\mathbb{C}^n)^*$, and let Φ be holomorphic near $\overline{\mathbb{D}}_R$ with values in $\text{Mat}_{n \times N}(\mathbb{C})$, satisfying

$$\Phi\Phi^* = \text{Id}_n \quad \text{on } \partial\mathbb{D}_R, \quad \Phi(0) = 0, \quad L\Phi'(0) = 0.$$

The boundary identity forces $N \geq n$. Choose $Q \in U(n)$ with first row $L/\|L\|$, where $\|L\|$ is the Euclidean dual norm. Multiplication by Q preserves that identity and the Hilbert–Schmidt Dirichlet integral. The rows r_i of $Q\Phi$ have norm one on $\partial\mathbb{D}_R$ and vanish to orders at least $m_1 = 2$ and $m_i = 1$ for $i \geq 2$. Writing $r_i(z) = \sum_{k \geq m_i} a_{ik}(z/R)^k$, with $a_{ik} \in \mathbb{C}^N$, Parseval’s identity and radial integration give

$$\sum_{k \geq m_i} \|a_{ik}\|^2 = 1, \quad \int_{\mathbb{D}_R} |r'_i|^2 dA = \pi \sum_{k \geq m_i} k \|a_{ik}\|^2 \geq \pi m_i.$$

Summing proves the lower bound $\pi(n + 1)$. It is attained by $U \text{diag}((z/R)^2, z/R, \dots, z/R)$ when the last $n - 1$ columns of $U \in U(n)$ span $\ker L$, appending $N - n$ zero columns if $N > n$. This multiplier optimality leaves the holomorphic realization and its remainder estimates to the next section.

4. UNIFORM HOLOMORPHIC REALIZATION AND DISK OPERATIONS

We realize the trace calculation by holomorphic families and then prove the two disk operations: uniform descent and scalar enlargement. Holomorphic mapping spaces and Stein neighborhoods of graphs provide a qualitative deformation theory [17]. Here the parameter radius and cubic area remainder must remain uniform as the disk and its source radius vary. We obtain this control by lifting to a fixed Stein submanifold of affine space, perturbing there, and projecting back to X .

Assume in this section that X is projective. Choose a projective embedding $\iota : X \hookrightarrow \mathbb{P}^{m-1}$; it will be used to represent $\iota \circ f$ by homogeneous coordinates and to construct the lift space $\mathcal{Y}$. The embedding ι is kept fixed as f and R vary, so the constants below may depend on ι .

We use the standard L^p and C^k norms on source regions, with Euclidean area measure. Vector and operator norms are induced by the fixed Hermitian and Euclidean metrics; matrix-valued L^2 estimates use the Hilbert–Schmidt norm.

4.1. Frame and lifting estimates

Uniform realization cannot be obtained by declaring the classes with fixed upper bounds on Area_R and B_R compact up to the source boundary. For $0 < r < 1$ and $\varepsilon > 0$, the functions $h_r : \mathbb{D} \rightarrow \mathbb{C}$ given by

$$h_r(z) := \varepsilon \frac{1 - r^2}{1 - rz}$$

are holomorphic near $\overline{\mathbb{D}}$ and satisfy

$$\int_{\mathbb{D}} |h'_r|^2 dA = \pi \varepsilon^2 r^2, \quad B_1(h_r) \leq \frac{\varepsilon}{2}, \quad |h'_r(1)| = \varepsilon \frac{r(1 + r)}{1 - r} \longrightarrow \infty.$$

Thus bounded area and bounded B_1 do not imply relative compactness in $C^1(\overline{\mathbb{D}})$ (with the Euclidean metric on $\mathbb{C}$); the uniform estimates below must be proved directly.

Boundary spectral factorization and a Green potential bound control a normalized frame. Homogeneous coordinates and Nicolau's corona theorem then give a uniformly bounded affine lift with bounded Dirichlet integral.

Fix $A_0, B_0 \geq 0$. In the auxiliary results below, f is holomorphic near $\overline{\mathbb{D}}_R$ and satisfies $\text{Area}_R(f) \leq A_0$ and $B_R(f) \leq B_0$. Norms on fixed finite-rank bundles are chosen once and for all.

The following factorization makes the frame in Theorem 4.7 unitary on the boundary, as required by the Ricci-trace identity.

Lemma 4.1. *Let $H : \partial\mathbb{D} \rightarrow \text{Mat}_{n \times n}(\mathbb{C})$ be a smooth positive-definite Hermitian matrix function. There is a map $A : \mathbb{D} \rightarrow \text{GL}_n(\mathbb{C})$, holomorphic in $\mathbb{D}$ and smooth on $\overline{\mathbb{D}}$ with invertible boundary values, such that*

$$A(\zeta)^* H(\zeta) A(\zeta) = \text{Id}_n \quad (|\zeta| = 1).$$

Proof. The Wiener–Masani theorem in the smooth boundary form of [3, Theorem 2.1] gives $H = h^* h$, where h and h^{-1} are holomorphic on $\mathbb{D}$ and smooth on $\overline{\mathbb{D}}$. Thus $A := h^{-1}$ has the required boundary normalization. $\square$

For $z \in \mathbb{D}$, let δ_z be the unit point mass at z . The Dirichlet Green function of the unit disk is

$$g(z, w) = \log \left| \frac{1 - \bar{z}w}{z - w} \right|, \quad -\Delta_w g = 2\pi\delta_z.$$

On $\mathbb{D}_R$ we use $g_R(z, w) := g(z/R, w/R)$.

The Green potential controls both the frame and the homogeneous lift. The weighted derivative bound controls its logarithmic singularity.

Lemma 4.2. *Let $\varepsilon_f^\omega(w) := |f_z(w)|_\omega^2$ be the area density of f , so that $f^*\omega = \varepsilon_f^\omega dA$ and $\text{Area}_R(f) = \int_{\mathbb{D}_R} \varepsilon_f^\omega dA$. If $\text{Area}_R(f) \leq A_0$ and $B_R(f) \leq B_0$, then*

$$\sup_{z \in \mathbb{D}_R} \int_{\mathbb{D}_R} g_R(z, w) \varepsilon_f^\omega(w) dA(w) \leq C_g(A_0, B_0) := C_G(B_0^2 + A_0), \quad (35)$$

where C_G is a universal constant, independent of R and f .

Proof. First reduce to the unit disk. Under the dilation $h(\zeta) := f(R\zeta)$, the identities

$$\text{Area}_1(h) = \text{Area}_R(f), \quad B_1(h) = B_R(f), \quad \varepsilon_h^\omega(\eta) = R^2 \varepsilon_f^\omega(R\eta)$$

show that the hypotheses are unchanged. Since $g_R(R\zeta, R\eta) = g(\zeta, \eta)$, change of variables also preserves the Green-potential integral. We may therefore assume $R = 1$.

Fix $z \in \mathbb{D}$ and set $\delta := 1 - |z|$, its distance to the boundary. Split the integral into the disk $|w - z| < \delta/2$ and its complement in $\mathbb{D}$. On the first region,

$$1 - |w| \geq 1 - |z| - |w - z| > \frac{\delta}{2}, \quad |f_z(w)|_\omega \leq \frac{B_0}{1 - |w|}.$$

Thus $\varepsilon_f^\omega(w) \leq 4B_0^2/\delta^2$. To bound the Green kernel, write

$$1 - \bar{z}w = (1 - |z|^2) + \bar{z}(z - w).$$

Since $1 - |z|^2 = (1 + |z|)\delta \leq 2\delta$, the triangle inequality gives

$$|1 - \bar{z}w| \leq 2\delta + |z - w| \leq \frac{5\delta}{2}, \quad g(z, w) \leq \log \frac{5\delta}{2|z - w|}.$$

Integration in polar coordinates bounds this contribution by

$$\frac{4B_0^2}{\delta^2} 2\pi \int_0^{\delta/2} r \log \frac{5\delta}{2r} dr = \pi B_0^2 \left(\log 5 + \frac{1}{2} \right).$$

On the complementary region $|z - w| \geq \delta/2$, the same triangle inequality instead yields

$$\frac{|1 - \bar{z}w|}{|z - w|} \leq 1 + \frac{2\delta}{|z - w|} \leq 5.$$

Thus $g(z, w) \leq \log 5$ there, and the remaining contribution is at most

$$(\log 5) \int_{\mathbb{D}} \varepsilon_f^\omega dA \leq A_0 \log 5.$$

Adding the two bounds proves (35); one may take the universal constant $C_G := \pi(\log 5 + 1/2)$, independently of z , R , and f . $\square$

We specify the curvature convention in local holomorphic coordinates $w^1, \dots, w^n$. Recall that h is the Gram matrix in (1), with squared norms c^*hc . For a section represented by a coefficient column c , the Chern connection and its curvature are

$$\nabla' c = \partial c + h^{-1}(\partial h)c, \quad \nabla'' c = \bar{\partial} c, \quad F^\nabla = \bar{\partial}(h^{-1}\partial h).$$

The coefficient of $dw^\alpha \wedge d\bar{w}^\beta$ in F^∇ is $-\partial_{\bar{w}^\beta}(h^{-1}\partial_{w^\alpha}h)$; the minus sign comes from reversing the order of the two differentials. For $v = \sum_\alpha v^\alpha \partial_{w^\alpha}$ and a tangent vector u with coefficient column c , our curvature tensor is

$$R^\omega(v, \bar{v}, u, \bar{u}) := \sum_{\alpha, \beta} v^\alpha \bar{v}^\beta c^* [-\partial_{\bar{w}^\beta} \partial_{w^\alpha} h + (\partial_{\bar{w}^\beta} h) h^{-1} (\partial_{w^\alpha} h)] c. \quad (36)$$

Thus positive holomorphic bisectional curvature means that (36) is positive for every pair of nonzero tangent vectors v, u at the same point. For a holomorphic disk f , the curvature endomorphism $\mathcal{R}_f^\omega$ is characterized by

$$\langle \mathcal{R}_f^\omega v, v \rangle_\omega = R^\omega(f_z, \bar{f}_z, v, \bar{v}). \quad (37)$$

With the normalizations in (1), a holomorphic section s of f^*T_X satisfies

$$\partial_z \partial_{\bar{z}} |s|_\omega^2 = |\nabla_z s|_\omega^2 - \langle \mathcal{R}_f^\omega s, s \rangle_\omega, \quad f^* \text{Ric}(\omega) = 2 \text{tr}(\mathcal{R}_f^\omega) dA. \quad (38)$$

Choose C_ω so that

$$\|\mathcal{R}_f^\omega(z)\|_{\text{op}} \leq C_\omega \varepsilon_f^\omega(z).$$

Compactness of X gives such a constant.

For a frame $e = (e_1, \dots, e_n)$, write $G = (\langle e_j, e_k \rangle_\omega)$. Matrix inequalities refer to Hermitian quadratic forms. The Green bound and Bochner identity control G throughout the disk and bound the frame's covariant Dirichlet integral. Both estimates enter the lifted-field bounds in Theorem 4.7.

Proposition 4.3. *There are constants $\Gamma \geq 1$ and $D_1 < \infty$, depending only on (X, ω, A_0, B_0) , such that every holomorphic frame e of f^*T_X that is smooth on the closed disk and unitary on its boundary has Gram matrix G satisfying*

$$\Gamma^{-1} \text{Id}_n \leq G \leq \Gamma \text{Id}_n. \quad (39)$$

Moreover

$$\sum_{a=1}^n \int_{\mathbb{D}_R} |\nabla_z e_a|^2 dA \leq D_1. \quad (40)$$

In particular, these constants are independent of R , f , and the boundary-unitary frame.

Proof. Berndtsson and Rosay obtain two-sided metric bounds from a bounded curvature potential in [3, Proposition 2.3]. We give the Bochner–Green calculation in terms of A_0, B_0 , including the covariant Dirichlet estimate (40). Set

$$u_G := \text{tr } G = \sum_{a=1}^n |e_a|_\omega^2 > 0, \quad q := \sum_{a=1}^n |\nabla_z e_a|_\omega^2.$$

The curvature bound and (38) give

$$\begin{aligned}\partial_z \partial_{\bar{z}} u_G &= q - \sum_a \langle \mathcal{R}_f^\omega e_a, e_a \rangle_\omega \\ &\geq q - C_\omega \varepsilon_f^\omega \sum_a |e_a|_\omega^2 = q - C_\omega \varepsilon_f^\omega u_G.\end{aligned}$$

Because e_a is holomorphic and the connection preserves the metric, $\partial_z u_G = \sum_a \langle e_a, \nabla_z e_a \rangle_\omega$. Cauchy–Schwarz, applied to the direct sum of the n fibers, gives

$$|\partial_z u_G|^2 \leq \left(\sum_a |e_a|_\omega^2 \right) \left(\sum_a |\nabla_z e_a|_\omega^2 \right) = u_G q.$$

Since $u_G > 0$, logarithmic differentiation now yields

$$\begin{aligned}\partial_z \partial_{\bar{z}} \log u_G &= \frac{\partial_z \partial_{\bar{z}} u_G}{u_G} - \frac{|\partial_z u_G|^2}{u_G^2} \\ &\geq \frac{q}{u_G} - C_\omega \varepsilon_f^\omega - \frac{q}{u_G} = -C_\omega \varepsilon_f^\omega.\end{aligned}$$

Hence

$$\Delta \log u_G \geq -4C_\omega \varepsilon_f^\omega, \quad u_G|_{\partial \mathbb{D}_R} = n.$$

Green representation, with $-\Delta_w g_R(z, w) = 2\pi \delta_z$, reads

$$\log \frac{u_G(z)}{n} = -\frac{1}{2\pi} \int_{\mathbb{D}_R} g_R(z, w) \Delta_w \log u_G(w) dA(w).$$

Since $g_R \geq 0$, Lemma 4.2 yields

$$\log \frac{u_G(z)}{n} \leq \frac{2C_\omega}{\pi} \int_{\mathbb{D}_R} g_R(z, w) \varepsilon_f^\omega(w) dA(w) \leq \frac{2C_\omega C_g(A_0, B_0)}{\pi}.$$

The algebraic dual holomorphic frame $e^a(e_b) = \delta_b^a$ has Gram matrix $G^{-\mathsf{T}} := (G^{-1})^\mathsf{T}$, where T denotes ordinary transpose. The curvature of the dual bundle is the negative transpose of the original curvature, so its operator norm is bounded by $C_\omega \varepsilon_f^\omega$. Applying the logarithmic trace estimate to the dual frame gives

$$\mathrm{tr} G \leq \Gamma, \quad \mathrm{tr}(G^{-1}) \leq \Gamma, \quad \Gamma := n \exp\left(\frac{2C_\omega C_g(A_0, B_0)}{\pi}\right).$$

For every eigenvalue $\mu > 0$ of G , $\mu \leq \mathrm{tr} G \leq \Gamma$ and $\mu^{-1} \leq \mathrm{tr}(G^{-1}) \leq \Gamma$. These two inequalities give (39).

Summing the Bochner identity gives

$$\partial_z \partial_{\bar{z}} \mathrm{tr} G = \sum_a |\nabla_z e_a|^2 - \mathrm{tr}(\mathcal{R}_f^\omega \mathsf{S}), \quad \mathsf{S} := \sum_a e_a \otimes e_a^\flat.$$

Here $e_a^\flat(v) = \langle e_a, v \rangle_\omega$ denotes metric duality. Since $G = \mathrm{Id}_n$ on $\partial \mathbb{D}_R$, Jacobi's formula gives

$$\partial_\nu \log \det G = \mathrm{tr}(G^{-1} \partial_\nu G) = \partial_\nu \mathrm{tr} G \quad \text{on } \partial \mathbb{D}_R.$$

Consequently Green's formula and (1) give

$$\int_{\mathbb{D}_R} \partial_z \partial_{\bar{z}} \mathrm{tr} G dA = \frac{1}{4} \int_{\partial \mathbb{D}_R} \partial_\nu \log \det G ds = -\frac{1}{2} \int_{\mathbb{D}_R} f^* \mathrm{Ric}(\omega).$$

Integrating the Bochner identity and solving for the nonnegative derivative term therefore gives

$$\sum_a \int_{\mathbb{D}_R} |\nabla_z e_a|^2 dA = \int_{\mathbb{D}_R} \mathrm{tr}(\mathcal{R}_f^\omega \mathsf{S}) dA - \frac{1}{2} \int_{\mathbb{D}_R} f^* \mathrm{Ric}(\omega).$$

Now $\text{tr } \mathbf{S} \leq n\Gamma$, while the absolute Ricci density is at most $2nC_\omega\varepsilon_f^\omega$, since $f^*\text{Ric} = 2\text{tr}(\mathcal{R}_f^\omega) dA$. Taking absolute values and using $\text{Area}_R(f) \leq A_0$ gives

$$\sum_a \int_{\mathbb{D}_R} |\nabla_z e_a|^2 dA \leq nC_\omega(\Gamma + 1)A_0.$$

Thus $D_1 := nC_\omega(\Gamma + 1)A_0$ is an admissible choice in (40). $\square$

Recall the projective embedding $\iota : X \hookrightarrow \mathbb{P}^{m-1}$ chosen at the beginning of this section. Let ω_{FS} be the Fubini–Study form on its target, with normalization

$$\omega_{\text{FS}} = \frac{i}{2} \partial \bar{\partial} \log \sum_{j=1}^m |Z_j|^2.$$

Choose $c_\iota > 0$ such that $\iota^*\omega_{\text{FS}} \leq c_\iota\omega$; such a constant exists by compactness of X . The displayed Fubini–Study formula is read in homogeneous coordinates $Z = (Z_1, \dots, Z_m) \neq 0$; it is unchanged as a form on projective space when the homogeneous representative is multiplied by a nonvanishing holomorphic function.

We choose homogeneous coordinates with a uniform lower bound for the corona theorem and a Dirichlet bound for the affine lift.

Lemma 4.4. *Every disk satisfying $\text{Area}_R(f) \leq A_0$ and $B_R(f) \leq B_0$ has a homogeneous lift $\mathbf{s}$ from an open neighborhood of $\overline{\mathbb{D}}_R$ to $\mathbb{C}^m \setminus \{0\}$, holomorphic on that neighborhood, with $[\mathbf{s}] = \iota \circ f$ and $|\mathbf{s}| = 1$ on the boundary. It satisfies*

$$|\mathbf{s}| \leq 1, \quad \int_{\mathbb{D}_R} |\mathbf{s}'|^2 dA = \int_{\mathbb{D}_R} f^* \iota^* \omega_{\text{FS}} \leq c_\iota A_0, \quad (41)$$

and

$$|\mathbf{s}(z)| \geq \delta_{\mathbf{s}} > 0, \quad (42)$$

where $\delta_{\mathbf{s}} = \delta_{\mathbf{s}}(X, \omega, \iota, A_0, B_0)$ is independent of R and f .

Proof. The tautological line bundle on $\mathbb{P}^{m-1}$ has fiber $\mathbb{C}Z$ over $[Z]$. Trivialize its pullback by $\iota \circ f$ on a slightly larger disk and choose a nonvanishing section, represented by a lift v in $\mathbb{C}^m \setminus \{0\}$. The positive real-analytic boundary function $|v|$ has a logarithm whose harmonic extension is the real part of a holomorphic function a on a slightly larger disk. Then $\mathbf{s} := e^{-a}v$ has boundary norm one, and the maximum principle gives $|\mathbf{s}| \leq 1$.

The area density of $\iota \circ f$ for the Fubini–Study metric is $\varepsilon_f^{\text{FS}} := |(\iota \circ f)'|_{\text{FS}}^2$, so that $f^* \iota^* \omega_{\text{FS}} = \varepsilon_f^{\text{FS}} dA$. Set $u := \log |\mathbf{s}|^2$. Since $\mathbf{s}$ has no common zero, $\Delta u = 4\varepsilon_f^{\text{FS}}$ and $u = 0$ on the boundary. Also $\Delta |\mathbf{s}|^2 = 4|\mathbf{s}'|^2$, and the normal derivatives of $|\mathbf{s}|^2$ and u agree where $|\mathbf{s}| = 1$. Hence

$$\int_{\mathbb{D}_R} |\mathbf{s}'|^2 dA = \frac{1}{4} \int_{\partial \mathbb{D}_R} \partial_\nu |\mathbf{s}|^2 ds = \frac{1}{4} \int_{\partial \mathbb{D}_R} \partial_\nu u ds = \int_{\mathbb{D}_R} \varepsilon_f^{\text{FS}} dA,$$

which proves (41). Green representation for u with zero boundary values gives

$$-u(z) = \frac{2}{\pi} \int g_R(z, w) \varepsilon_f^{\text{FS}}(w) dA(w) \leq \frac{2c_\iota C_g(A_0, B_0)}{\pi}.$$

Thus (42) holds with $\delta_{\mathbf{s}} = \exp(-c_\iota C_g(A_0, B_0)/\pi)$. $\square$

Write $H^\infty(\mathbb{D})$ for the bounded holomorphic functions with the supremum norm, and let

$$\mathcal{D}(\mathbb{D}) := \left\{ h : \mathbb{D} \rightarrow \mathbb{C} \text{ holomorphic} : \int_{\mathbb{D}} |h'|^2 dA < \infty \right\}.$$

The Dirichlet algebra is

$$\mathfrak{A} := H^\infty(\mathbb{D}) \cap \mathcal{D}(\mathbb{D}), \quad \|h\|_{\mathfrak{A}} := \|h\|_\infty + \left(\int_{\mathbb{D}} |h'|^2 dA \right)^{1/2}.$$

The corona estimate makes the affine lift uniform in the disk. Nicolau's theorem and the bound at the beginning of its proof [41, pp. 135–136] give the following statement.

Theorem 4.5. *For every $m \geq 1$ and $A, \delta > 0$ there is $C_N(m, A, \delta) < \infty$ such that, whenever $u_1, \dots, u_m \in \mathfrak{A}$ satisfy*

$$\|u_j\|_{\mathfrak{A}} \leq A, \quad \max_j |u_j(z)| \geq \delta \quad (z \in \mathbb{D}),$$

there are $v_1, \dots, v_m \in \mathfrak{A}$ with

$$\sum_j u_j v_j = 1, \quad \|v_j\|_{\mathfrak{A}} \leq C_N(m, A, \delta).$$

The same statement holds on every disk after dilation.

Proof. The case $\delta > A$ is vacuous; assume $\delta \leq A$. For $p = 2$, Nicolau uses the norm

$$\|h\|_{\text{Nic}} := \|h\|_{\infty} + \left(\frac{1}{\pi} \int_{\mathbb{D}} |h'|^2 dA \right)^{1/2}, \quad \|h\|_{\text{Nic}} \leq \|h\|_{\mathfrak{A}} \leq \sqrt{\pi} \|h\|_{\text{Nic}}.$$

His corona condition is the same maximum lower bound as above [41, p. 135, (1)]. The generators u_j/A therefore have Nicolau norm at most 1 and lower bound δ/A . The estimate at the beginning of his proof [41, p. 136] gives solutions b_j with $\sum_j (u_j/A) b_j = 1$ and $\|b_j\|_{\text{Nic}} \leq K_m(\delta/A)$, where K_m depends only on the indicated data. Setting $v_j := b_j/A$ gives $\sum_j u_j v_j = 1$ and

$$\|v_j\|_{\mathfrak{A}} \leq \frac{\sqrt{\pi}}{A} K_m(\delta/A),$$

which is an admissible choice of $C_N(m, A, \delta)$. For a disk of radius R ,

$$\|h(R \cdot)\|_{\infty, \mathbb{D}} = \|h\|_{\infty, \mathbb{D}_R}, \quad \int_{\mathbb{D}} |(h(R\zeta))'|^2 dA(\zeta) = \int_{\mathbb{D}_R} |h'(z)|^2 dA(z),$$

so the same bound applies after dilation. $\square$

For a normalized lift $\mathbf{s}$ from Lemma 4.4, its coordinates have algebra norm at most $C_{\mathbf{s}} := 1 + \sqrt{c_{\iota} A_0}$, and at least one has modulus at least $\delta_{\mathbf{s}}/\sqrt{m}$ at every point, because $|\mathbf{s}|^2 \leq m \max_j |\mathbf{s}_j|^2$. Theorem 4.5 gives a holomorphic map $\chi : \mathbb{D}_R \rightarrow (\mathbb{C}^m)^*$, whose components belong to the algebra on $\mathbb{D}_R$ obtained from $\mathfrak{A}$ by dilation, satisfying

$$\chi \cdot \mathbf{s} = 1, \quad \|\chi\|_{\infty} + \|\chi'\|_2 \leq C_{\chi} \tag{43}$$

for a constant $C_{\chi} = C_{\chi}(X, \omega, \iota, A_0, B_0)$. The dot denotes the bilinear pairing $\chi \cdot \mathbf{s} = \sum_{j=1}^m \chi_j \mathbf{s}_j$, without complex conjugation. The constant absorbs the fixed dimension factors in passing from scalar to vector norms.

To extend the lift past $\overline{\mathbb{D}}_R$, dilate χ radially while keeping $\mathbf{s}$ fixed, then restore their pairing by division. For $0 < \rho < 1$ sufficiently close to 1, set

$$\chi_{\rho}(z) := \chi(\rho z), \quad \gamma_{\rho} := \chi_{\rho} \cdot \mathbf{s}, \quad \hat{\chi} := \frac{\chi_{\rho}}{\gamma_{\rho}}.$$

Indeed, $\gamma_{\rho} - 1 = \chi(\rho \cdot) \cdot (\mathbf{s} - \mathbf{s}(\rho \cdot)) \rightarrow 0$ uniformly on $\overline{\mathbb{D}}_R$, so we may require $|\gamma_{\rho}| \geq 1/2$ there. The quotient is therefore holomorphic near $\overline{\mathbb{D}}_R$ and satisfies $\hat{\chi} \cdot \mathbf{s} = 1$ and $\|\hat{\chi}\|_{\infty} \leq 2C_{\chi}$. Dilation gives $\|\chi'_{\rho}\|_2 \leq C_{\chi}$; the product and quotient rules give

$$\begin{aligned} \|\gamma'_{\rho}\|_2 &\leq C_{\chi}(1 + \sqrt{c_{\iota} A_0}), \\ \|\hat{\chi}'\|_2 &\leq 2C_{\chi} + 4C_{\chi}^2(1 + \sqrt{c_{\iota} A_0}) =: D_{\chi}. \end{aligned} \tag{44}$$

All norms here are on $\mathbb{D}_R$. The bound depends only on the fixed target, embedding, and A_0, B_0 ; the base lift $\mathbf{s}$ is unchanged.

4.2. Uniform lifts and holomorphic variations

The pairs $(\mathbf{s}, \chi)$ lie in a fixed affine model: the equation $\chi \cdot \mathbf{s} = 1$ excludes the cone vertex. A holomorphic right inverse lifts tangent fields to this model. The model depends only on the embedding; the area and weighted-derivative caps determine a compact subset and a retraction near it.

Lemma 4.6. *For the projective embedding $\iota : X \hookrightarrow \mathbb{P}^{m-1}$ chosen above, there are a closed Stein submanifold $\mathcal{Y} \subset \mathbb{C}^{2m}$, a holomorphic submersion $\pi_X : \mathcal{Y} \rightarrow X$, and a holomorphic bundle map $\mathcal{L} : \pi_X^* T_X \rightarrow T_{\mathcal{Y}}$ such that $d\pi_X \circ \mathcal{L} = \text{Id}$. These objects depend only on the fixed embedding.*

Proof. Consider the affine cone of the embedding,

$$\widehat{X} := \{0\} \cup \{\mathbf{s} \in \mathbb{C}^m \setminus \{0\} : [\mathbf{s}] \in \iota(X)\},$$

and set

$$\mathcal{Y} := \{(\mathbf{s}, \chi) \in \widehat{X} \times (\mathbb{C}^m)^* : \chi \cdot \mathbf{s} = 1\}, \quad \pi_X(\mathbf{s}, \chi) := \iota^{-1}[\mathbf{s}]. \quad (45)$$

The defining equation excludes the cone vertex. On a chart $\mathbf{s}_j \neq 0$, local coordinates consist of coordinates on X , a nonzero scale factor, and $m - 1$ unconstrained dual coordinates. Hence $\mathcal{Y}$ is a smooth closed affine algebraic variety and therefore Stein, while π_X is a holomorphic submersion.

For the holomorphic splitting, set $V := \mathbb{C}^m$ and let $\ell := \iota^* \mathcal{O}_{\mathbb{P}^{m-1}}(-1)$ be the tautological line bundle on X ; its fiber $\ell_x \subset V$ is the line representing $\iota(x)$. Define

$$\mathcal{U} := \{(x, \chi) \in X \times V^* : \chi|_{\ell_x} \neq 0\}.$$

The map

$$\Theta : \mathcal{Y} \longrightarrow \mathcal{U}, \quad (\mathbf{s}, \chi) \longmapsto (\pi_X(\mathbf{s}, \chi), \chi),$$

is biholomorphic. Indeed, if s_0 is a local nonvanishing holomorphic section of ℓ , then

$$\sigma(x, \chi) := \frac{s_0(x)}{\chi(s_0(x))}$$

is independent of the choice of s_0 , is holomorphic, lies in ℓ_x , and satisfies $\chi(\sigma(x, \chi)) = 1$. Thus

$$\Theta^{-1}(x, \chi) = (\sigma(x, \chi), \chi).$$

Since $\mathcal{U}$ is open in $X \times V^*$, its tangent space at (x, χ) is $T_x X \oplus V^*$. Transporting the vectors $(v, 0)$ by $d\Theta^{-1}$ gives the required holomorphic right inverse

$$\mathcal{L} : \pi_X^* T_X \rightarrow T_{\mathcal{Y}}, \quad \mathcal{L}_y(v) := d(\Theta^{-1})_{\Theta(y)}(v, 0), \quad d\pi_X \circ \mathcal{L} = \text{Id}. \quad (46)$$

□

The normalized coordinates and their corona partners place every disk in a common compact subset of the fixed affine model. The frame estimates then control its lifted tangent fields. We assemble these data before constructing the holomorphic variations.

Theorem 4.7. *Fix the affine model $\mathcal{Y}$, the submersion π_X , and the right inverse $\mathcal{L}$ supplied by Lemma 4.6. Given $A_0, B_0 \geq 0$, one can choose a compact set $K_{A_0, B_0} \Subset \mathcal{Y}$, an open neighborhood Ω of K_{A_0, B_0} in $\mathbb{C}^{2m}$, a holomorphic map $\tilde{\pi}_X : \Omega \rightarrow X$ with $\tilde{\pi}_X|_{\mathcal{Y} \cap \Omega} = \pi_X$, and constants*

$$\Gamma, D_1, D_{\tilde{f}}, C_{\theta, \infty}, D_{\theta} < \infty.$$

These cap-dependent choices depend only on $(X, \omega, \iota, A_0, B_0)$ and are independent of the source radius and the individual disk; the affine model and its splitting depend only on ι . For every $R > 0$ and every map f holomorphic on a neighborhood of $\mathbb{D}_R$ with $\text{Area}_R(f) \leq A_0$ and $B_R(f) \leq B_0$, one can choose a frame e , a lifted disk $\tilde{f}$, and a matrix θ compatibly so that the following properties hold:

- (i) e is a holomorphic frame of f^*T_X , smooth on the closed disk and unitary on its boundary, whose Gram matrix $G(z) := (\langle e_j(z), e_k(z) \rangle_\omega)_{j,k=1}^n$ satisfies

$$\Gamma^{-1} \text{Id}_n \leq G \leq \Gamma \text{Id}_n, \quad \sum_{a=1}^n \int_{\mathbb{D}_R} |\nabla_z e_a|^2 dA \leq D_1;$$

- (ii) $\tilde{f}$ is a map from a neighborhood of $\overline{\mathbb{D}_R}$ to $\mathcal{Y}$, holomorphic on that neighborhood, such that

$$\pi_X \circ \tilde{f} = f, \quad \tilde{f}(\overline{\mathbb{D}_R}) \subset K_{A_0, B_0}, \quad \int_{\mathbb{D}_R} |\tilde{f}'|^2 dA \leq D_{\tilde{f}};$$

- (iii) the frame vectors have holomorphic lifts $\theta_1, \dots, \theta_n : \mathbb{D}_R \rightarrow \mathbb{C}^{2m}$ along $\tilde{f}$. Their column matrix and the lifting identities are

$$\begin{aligned} \theta &:= (\theta_1, \dots, \theta_n) : \mathbb{D}_R \rightarrow \text{Mat}_{2m \times n}(\mathbb{C}), \\ d(\tilde{\pi}_X)_{\tilde{f}(z)} \theta_a(z) &= e_a(z) \quad (1 \leq a \leq n). \end{aligned} \tag{47}$$

The matrix θ is smooth on the closed disk and satisfies

$$\sup_{z \in \mathbb{D}_R} \|\theta(z)\|_{\text{op}} \leq C_{\theta, \infty}, \quad \int_{\mathbb{D}_R} \|\theta'\|_{\text{HS}}^2 dA \leq D_\theta.$$

Proof. Fix a disk f satisfying the caps A_0, B_0 .

- (i) Choose $R' > R$ so that $\overline{\mathbb{D}_{R'}}$ lies in the given source neighborhood, and choose a holomorphic frame $\tilde{e}$ of f^*T_X on $\overline{\mathbb{D}_{R'}}$; such a frame exists by the Oka–Grauert principle [23, Satz 6, p. 270], since the source is contractible and Stein. Restrict its Gram matrix to $\partial \mathbb{D}_R$, rescale by $z = R\zeta$, and apply Lemma 4.1. If $A(\zeta)$ is the resulting factor for the rescaled Gram matrix, set $e(z) := \tilde{e}(z)A(z/R)$. Its boundary Gram matrix is $A(z/R)^* \tilde{G}(z) A(z/R) = \text{Id}_n$. Thus e is holomorphic, smooth on the closed disk, and unitary on its boundary. Proposition 4.3 gives (39) and (40), proving (i).

- (ii) By Lemma 4.4, f has a homogeneous lift $\mathbf{s}$ with

$$|\mathbf{s}| \leq 1, \quad |\mathbf{s}| \geq \delta_{\mathbf{s}}, \quad \int_{\mathbb{D}_R} |\mathbf{s}'|^2 dA \leq c_l A_0.$$

Its coordinates satisfy the hypotheses of Theorem 4.5 with algebra-norm bound $C_{\mathbf{s}}$ and corona lower bound $\delta_{\mathbf{s}}/\sqrt{m}$. Thus (43) supplies a vector χ satisfying $\chi \cdot \mathbf{s} = 1$, and the dilation construction leading to (44) gives a vector $\hat{\chi}$, holomorphic near the closed disk, with

$$\hat{\chi} \cdot \mathbf{s} = 1, \quad \|\hat{\chi}\|_{L^\infty(\mathbb{D}_R)} \leq 2C_\chi, \quad \|\hat{\chi}'\|_{L^2(\mathbb{D}_R)} \leq D_\chi.$$

The identity $\hat{\chi} \cdot \mathbf{s} = 1$ shows that $\tilde{f} := (\mathbf{s}, \hat{\chi})$ takes values in the fixed space $\mathcal{Y}$ of (45). It satisfies

$$\tilde{f}(\overline{\mathbb{D}_R}) \subset K_{A_0, B_0} := \mathcal{Y} \cap \{|\mathbf{s}| \leq 1, |\chi| \leq 2C_\chi\}, \quad \int_{\mathbb{D}_R} |\tilde{f}'|^2 dA \leq c_l A_0 + D_\chi^2 =: D_{\tilde{f}}. \tag{48}$$

The set K_{A_0, B_0} is compact: it is closed and bounded, and $\chi \cdot \mathbf{s} = 1$ prevents $\mathbf{s}$ from approaching the cone vertex.

Since $\mathcal{Y}$ is a closed Stein submanifold of $\mathbb{C}^{2m}$, the Docquier–Grauert tubular neighborhood theorem [13, Satz 3], in the holomorphic deformation-retraction formulation [18, Theorem 3.1], gives a holomorphic retraction $\mathbf{r} : \Omega \rightarrow \mathcal{Y} \cap \Omega$ from a neighborhood Ω of K_{A_0, B_0} , equal to the identity on $\mathcal{Y} \cap \Omega$. Its derivatives are bounded on smaller compact neighborhoods of K_{A_0, B_0} . Set $\tilde{\pi}_X := \pi_X \circ \mathbf{r}$. Since $\pi_X \circ \tilde{f} = f$ and $\tilde{\pi}_X|_{\mathcal{Y} \cap \Omega} = \pi_X$, (48) proves (ii). The choices of K_{A_0, B_0} , Ω , and $\tilde{\pi}_X$ depend only on the fixed target data and the two caps; they are fixed before any individual lifted disk $\tilde{f}$ is chosen. The space $\mathcal{Y}$, the projection π_X , and the splitting $\mathcal{L}$ were already fixed from the embedding alone.

(iii) Regard the splitting $\mathcal{L}$ from (46) as a bundle map from $\pi_X^*T_X$ to the ambient trivial bundle $\mathcal{Y} \times \mathbb{C}^{2m}$. In $\nabla\mathcal{L}$, use the pullback Chern connection on $\pi_X^*T_X$ and the flat connection on this ambient bundle. Let

$$L_0 := \sup_{K_{A_0, B_0}} \|\mathcal{L}\|, \quad L_1 := \sup_{K_{A_0, B_0}} \|\nabla\mathcal{L}\|.$$

Set

$$\theta_a(z) := \mathcal{L}_{\tilde{f}(z)} e_a(z).$$

Then $d(\tilde{\pi}_X)_{\tilde{f}} \theta_a = e_a$. With the preceding connection convention, ambient differentiation gives

$$\theta'_a = \mathcal{L}_{\tilde{f}} \nabla_z e_a + (\nabla_{\tilde{f}} \mathcal{L}) e_a. \quad (49)$$

The frame and lifted-disk bounds, with $|a+b|^2 \leq 2|a|^2 + 2|b|^2$, give

$$\|\theta\|_{L^\infty, \text{op}} \leq L_0 \sqrt{\Gamma} =: C_{\theta, \infty}, \quad (50)$$

$$\int_{\mathbb{D}_R} \|\theta'\|_{\text{HS}}^2 dA \leq 2L_0^2 D_1 + 2n\Gamma L_1^2 D_{\tilde{f}} =: D_\theta. \quad (51)$$

This proves (iii). $\square$

We now add the lifted variation fields and project to X . Radial regularization extends the fields past the closed disk while keeping the lifted map fixed. The multiplier retains its vanishing orders, and the paired trace converges to the value computed in Section 3. The parameter range and cubic remainder remain uniform throughout this approximation.

The choices have a fixed order. The embedding determines $\mathcal{Y}, \pi_X, \mathcal{L}$, independently of the disks and caps; A_0, B_0 then determine $K_{A_0, B_0}, \Omega, \tilde{\pi}_X$. Together with the multiplier bounds D_Φ, N_Φ , these data determine t_0, C_T before R, f, Φ , or the regularization parameter ρ is chosen. Each disk has its own compatible frame, lift, and lifted fields, subject to the bounds of Theorem 4.7. The choice of ρ may depend on these individual data and on Φ ; for trace approximation it may also depend on the prescribed error ε . The constants t_0, C_T remain unchanged. The holomorphic extension neighborhood may depend on the individual family, Φ , and ρ ; no uniform extension width is asserted.

Theorem 4.8. *Fix (X, ω) , the projective embedding $\iota : X \hookrightarrow \mathbb{P}^{m-1}$, and the bounds $A_0, B_0 \geq 0$ as in Theorem 4.7. Let $D_\Phi \geq 0$, and let $N_\Phi \geq 1$ be an integer. There are constants $t_0 > 0$ and $C_T < \infty$, depending only on $(X, \omega, \iota, A_0, B_0, D_\Phi, N_\Phi)$, with the following property.*

Let $R > 0$, and let $f : \mathbb{D}_R \rightarrow X$ extend holomorphically to a neighborhood of $\overline{\mathbb{D}_R}$ and satisfy $\text{Area}_R(f) \leq A_0$ and $B_R(f) \leq B_0$. Use the fixed K_{A_0, B_0}, Ω , and $\tilde{\pi}_X$ from Theorem 4.7, and choose any compatible disk-dependent triple $e, \tilde{f}, \theta$ supplied by that theorem. Suppose that $\Phi = (\Phi_1, \dots, \Phi_{N_\Phi}) : \mathbb{D}_R \rightarrow \text{Mat}_{n \times N_\Phi}(\mathbb{C})$ extends holomorphically to an open neighborhood of $\overline{\mathbb{D}_R}$ and satisfies

$$\sup_{z \in \overline{\mathbb{D}_R}} \|\Phi(z)\|_{\text{op}} \leq 1, \quad \int_{\mathbb{D}_R} \|\Phi'\|_{\text{HS}}^2 dA \leq D_\Phi,$$

*where the column Φ_a determines the section $e\Phi_a = \sum_{j=1}^n e_j \Phi_{ja}$ of f^*T_X . The constants t_0 and C_T are independent of R, f, Φ and of this compatible triple.*

For each $0 < \rho < 1$, write

$$\Xi_\rho(z) := \theta(\rho z) \Phi(z), \quad (\Xi_\rho)_a(z) := \sum_{j=1}^n \theta_j(\rho z) \Phi_{ja}(z).$$

(i) Holomorphic realization. *For every $0 < \rho < 1$, each column defines the family*

$$f_{a,t}^\rho(z) := \tilde{\pi}_X(\tilde{f}(z) + t(\Xi_\rho)_a(z)), \quad t \in \mathbb{C}, \quad |t| \leq t_0, \quad 1 \leq a \leq N_\Phi.$$

All these families are jointly holomorphic on one open neighborhood of $\overline{\mathbb{D}}_R \times \{t \in \mathbb{C} : |t| \leq t_0\}$; for every $0 < \rho < 1$ that neighborhood may depend on the selected disk-dependent triple, Φ , and ρ , but not on the column a .

- (ii) Uniform cubic remainder. For every $0 < \rho < 1$, $1 \leq a \leq N_\Phi$, and $\tau \in \{1, i\}$, write $t = \tau s$ with $s \in \mathbb{R}$ and set

$$\ell_{a,\tau}^\rho := \frac{d}{ds} \Big|_{s=0} \text{Area}_R(f_{a,\tau s}^\rho), \quad H_{a,\tau}^\rho := \frac{d^2}{ds^2} \Big|_{s=0} \text{Area}_R(f_{a,\tau s}^\rho).$$

For $|s| \leq t_0$,

$$\text{Area}_R(f_{a,\tau s}^\rho) \leq \text{Area}_R(f) + \ell_{a,\tau}^\rho s + \frac{1}{2} H_{a,\tau}^\rho s^2 + C_T |s|^3 \quad (52)$$

with the same constant C_T for both choices of τ .

- (iii) Trace approximation. If Φ is boundary coisometric, meaning $\Phi\Phi^* = \text{Id}_n$ on $\partial\mathbb{D}_R$, then for every $\varepsilon > 0$ there is $0 < \rho_\varepsilon < 1$, depending only on the individual data displayed above and on ε , such that

$$\left| \sum_{a=1}^{N_\Phi} (H_{a,1}^{\rho_\varepsilon} + H_{a,i}^{\rho_\varepsilon}) - \left(4 \int_{\mathbb{D}_R} \|\Phi'\|_{\text{HS}}^2 dA - 2 \int_{\mathbb{D}_R} f^* \text{Ric}(\omega) \right) \right| \leq \varepsilon. \quad (53)$$

Only this conclusion requires choosing ρ_ε close to 1; t_0 and C_T remain independent of ε and ρ_ε .

Proof. Use the fixed cap-dependent ambient data and the selected compatible triple from Theorem 4.7. Set

$$\delta_0 := \min\{1, \text{dist}(K_{A_0, B_0}, \mathbb{C}^{2m} \setminus \Omega)\} > 0,$$

and choose a compact $K' \Subset \Omega$ whose interior contains the closed Euclidean $3\delta_0/4$ -thickening of K_{A_0, B_0} . In the standard coordinate frame of $\mathbb{C}^{2m}$, let g_0 be the positive-semidefinite Hermitian coefficient matrix of $\tilde{\pi}_X^* \omega$ on K' , so

$$v^* g_0(w) v = |d(\tilde{\pi}_X)_w(v)|_\omega^2$$

for a coefficient column v . Only smooth coefficient bounds are used; the pullback form may be degenerate along the fibers. Take $C_{g,3}$ to bound the coefficients and their derivatives through order three on K' .

Take a multiplier Φ satisfying the stated bounds and, for $0 < \rho < 1$, set $\Xi_\rho := \theta(\rho \cdot) \Phi$. Dilation gives $\int_{\mathbb{D}_R} |\rho \theta'(\rho z)|^2 dA(z) = \int_{\mathbb{D}_{\rho R}} |\theta'(w)|^2 dA(w) \leq D_\theta$. Use this identity, $\|\Phi\|_\infty \leq 1$, and $(\theta(\rho z) \Phi)' = \rho \theta'(\rho z) \Phi + \theta(\rho z) \Phi'$ to obtain

$$\sup_{z \in \mathbb{D}_R} \|\Xi_\rho(z)\|_{\text{op}} \leq C_{\theta, \infty}, \quad \int_{\mathbb{D}_R} \|\Xi'_\rho\|_{\text{HS}}^2 dA \leq 2D_\theta + 2C_{\theta, \infty}^2 D_\Phi =: D_\Xi. \quad (54)$$

Choose

$$t_0 := \min \left\{ 1, \frac{\delta_0}{4 \max\{1, C_{\theta, \infty}\}} \right\}.$$

For $|t| \leq 2t_0$, the displacement of each column perturbation is at most $2t_0 C_{\theta, \infty} \leq \delta_0/2$. Thus $\tilde{f} + t(\Xi_\rho)_a$ stays in K' , uniformly in the disk and the column.

- (i) Fix a column and write $\Xi := (\Xi_\rho)_a$. The function $\theta(\rho z)$ is holomorphic for $|z| < R/\rho$; since $\tilde{f}$ and Φ extend past the closed disk, so does $\tilde{f} + t\Xi$. On the closed disk and for $|t| \leq t_0$ its image lies in the closed $\delta_0/2$ -thickening of K_{A_0, B_0} , which is compactly contained in $\text{int } K' \Subset \Omega$. Compactness gives a neighborhood of $\mathbb{D}_R \times \{|t| \leq t_0\}$ on which the image remains in Ω , so the projected family is jointly holomorphic there. Intersect these neighborhoods over the finitely many columns.

(ii) In ambient coordinates write $w(t) := \tilde{f} + t\Xi$ and $v(t) := \tilde{f}' + t\Xi'$. The area density is $v(t)^*g_0(w(t))v(t)$. Along either real parameter axis $t = \tau s$, with $\tau \in \{1, i\}$, the functions $w(\tau s)$ and $v(\tau s)$ are affine in s , with first derivatives $\tau\Xi$ and $\tau\Xi'$. For fixed z , put $G(s) := g_0(w(\tau s))$, $V(s) := v(\tau s)$, and $Q := \tau\Xi'$. Ordinary differentiation in the real variable s gives

$$\frac{d^3}{ds^3}(V^*GV) = V^*G'''V + 3Q^*G''V + 3V^*G''Q + 6Q^*G'Q.$$

The chain rule bounds $|G^{(k)}(s)|$ by $C_{g,3}|\Xi|^k$ for $1 \leq k \leq 3$, up to fixed dimension factors. The bounds $\|\Xi\|_\infty \leq C_{\theta,\infty}$ and $|t| \leq t_0$ consequently give the integrable estimate

$$\left| \frac{d^3}{ds^3}(v(\tau s)^*g_0(w(\tau s))v(\tau s)) \right| \leq C(C_{g,3}, C_{\theta,\infty}, t_0)(|\tilde{f}'|^2 + |\Xi'|^2).$$

Integrating and applying (48) and (54) give a uniform constant C_T . Taylor's theorem with remainder then proves (52) simultaneously for the two real parameter axes.

(iii) The tangent fields of the actual families satisfy, for each fixed f ,

$$\xi_{\rho,a} := d(\tilde{\pi}_X)_{\tilde{f}}(\theta(\rho \cdot)\Phi_a) \longrightarrow e\Phi_a \quad \text{in } C^1(\overline{\mathbb{D}}_R) \quad (\rho \uparrow 1),$$

because $\theta(\rho \cdot) \rightarrow \theta$ in the same norm. If Φ is boundary coisometric, Lemma 3.3, continuity of (29), and Proposition 3.4 therefore give

$$\sum_a \mathcal{H}_{R,f}(\xi_{\rho,a}) \longrightarrow 4 \int_{\mathbb{D}_R} \|\Phi'\|_{\text{HS}}^2 dA - 2 \int_{\mathbb{D}_R} f^* \text{Ric}(\omega).$$

The summand is $H_{a,1}^\rho + H_{a,i}^\rho$ by Lemma 3.3. Choose ρ_ε close enough to one to obtain the prescribed error. Parts (i)–(ii) and (54) hold for every $\rho \in (0, 1)$ with the same t_0, C_T . Their dependence is only on the fixed target data and A_0, B_0, D_Φ, N_Φ . $\square$

4.3. A common area and derivative estimate

For the fixed caps, Theorem 4.7 supplies a common compact set K , ambient projection Π , and Dirichlet bound D . Each compatible lift H satisfies

$$\Pi H = f, \quad H(\overline{\mathbb{D}}_R) \subset K, \quad \int_{\mathbb{D}_R} |H'|^2 dA \leq D. \quad (55)$$

Choose $\epsilon > 0$ such that the closed Euclidean ϵ -neighborhood K_ϵ of K is compactly contained in Ω . Write $g(w)$ for the Hermitian coefficient matrix of $\Pi^*\omega$ in ambient coordinates. Thus

$$a^*g(w)a = |d\Pi_w(a)|_\omega^2.$$

The form can be degenerate. On K_ϵ , fix bounds

$$\|g(w)\|_{\text{op}} \leq M_0, \quad \|Dg(w)[v]\|_{\text{op}} \leq L_g|v|, \quad (56)$$

where Dg is the real derivative in ambient coordinates. These bounds are fixed before selecting an individual disk.

Lemma 4.9. *If $F : \mathbb{D}_R \rightarrow \mathbb{C}^N$ is holomorphic and $\int_{\mathbb{D}_R} |F'|^2 dA \leq E$, then*

$$(R - |z|)|F'(z)| \leq \sqrt{E/\pi} \quad (z \in \mathbb{D}_R).$$

Proof. For $0 < r < R - |z|$, the submean inequality for $|F'|^2$ gives

$$\pi r^2 |F'(z)|^2 \leq \int_{\mathbb{D}_r(z)} |F'|^2 dA \leq E.$$

Let $r \uparrow R - |z|$. $\square$

Proposition 4.10. *Suppose H satisfies (55), and let Ξ be holomorphic near $\overline{\mathbb{D}}_R$, with*

$$\|\Xi\|_\infty \leq S, \quad \int_{\mathbb{D}_R} |\Xi'|^2 dA \leq E.$$

Fix $0 < \bar{t} \leq 1$ with $\bar{t}S < \epsilon$. For $|t| \leq \bar{t}$, set $f_t = \Pi(H + t\Xi)$ and

$$C_A = L_g S D + 2M_0 \sqrt{DE} + M_0 \bar{t} E. \quad (57)$$

Then

$$|\text{Area}_R(f_t) - \text{Area}_R(f)| \leq C_A |t|, \quad (58)$$

$$|B_R(f_t)^2 - B_R(f)^2| \leq \frac{C_A}{\pi} |t|. \quad (59)$$

In particular, both an upper and a lower bound for the new B follow:

$$\sqrt{\max\{0, B_R(f)^2 - C_A |t|/\pi\}} \leq B_R(f_t) \leq \sqrt{B_R(f)^2 + C_A |t|/\pi}. \quad (60)$$

The same family has the pointwise displacement bound

$$\text{dist}_\omega(f_t(z), f(z)) \leq \sqrt{M_0} S |t| \quad (z \in \mathbb{D}_R). \quad (61)$$

Proof. The entire segment $H(z) + st\Xi(z)$, $0 \leq s \leq 1$, lies in K_ϵ , so (56) applies along it. Put $a = H'(z)$, $b = \Xi'(z)$, and $g_t = g(H(z) + t\Xi(z))$. The exact difference of area densities is

$$(a + tb)^* g_t(a + tb) - a^* g_0 a = a^* (g_t - g_0) a + 2 \text{Re}(ta^* g_t b) + |t|^2 b^* g_t b. \quad (62)$$

Its absolute value is at most

$$|t| L_g S |a|^2 + 2M_0 |t| |a| |b| + M_0 |t|^2 |b|^2.$$

Integrating, and applying Cauchy–Schwarz to the middle term, proves (58).

For the second estimate, put $d = R - |z|$. Lemma 4.9 gives $d|a| \leq \sqrt{D/\pi}$ and $d|b| \leq \sqrt{E/\pi}$. Multiplying the same pointwise estimate by d^2 therefore bounds the difference of the two squared weighted derivatives by $C_A |t|/\pi$. Use $|\sup p - \sup q| \leq \sup |p - q|$ and $B_R(f)^2 = \sup d^2 |f_z|_\omega^2$ to obtain (59). Finally, the path $s \mapsto \Pi(H(z) + st\Xi(z))$, $0 \leq s \leq 1$, has speed at most $\sqrt{M_0} |t| |\Xi(z)|$. Its length proves (61). Only bounds on the possibly degenerate metric coefficients have been used. $\square$

4.4. Uniform descent and enlargement

When $\text{Ric}(\omega) \geq \kappa \omega$ and $\text{Area}_R(f) > A_*$, the trace is negative. Selecting one real direction and the sign of its parameter gives descent; the common parameter radius and Taylor remainder give a uniform step size under fixed area and weighted-derivative bounds.

Proof of Proposition 2.1. Put $x_0 = f(0)$ and $\lambda = \lambda_{x_0}^F$. We allow the scalar $k = \lambda(f_z(0))$ to be arbitrary; its value is preserved below.

For the multiplier (32), one has $N_\Phi = n$, $D_\Phi = \pi(n+1)$, and, by (34),

$$4\pi(n+1) - 2\kappa \text{Area}_R(f) \leq -2\kappa \delta.$$

Apply Theorem 4.8 with trace error $\varepsilon = \kappa \delta$, choose the value ρ_ε supplied by part (iii), and abbreviate $f_{a,t} = f_{a,t}^{\rho_\varepsilon}$, $\ell_{a,\tau} = \ell_{a,\tau}^{\rho_\varepsilon}$, and $H_{a,\tau} = H_{a,\tau}^{\rho_\varepsilon}$. For every column a and every $|t| \leq t_0$, the identities $\Phi_a(0) = 0$ and $d(\tilde{\pi}_X)_{\tilde{f}(0)} \theta(0) = e(0)$ give the exact interpolation formula

$$\begin{aligned} f_{a,t}(0) &= f(0), \\ (f_{a,t})_z(0) &= f_z(0) + t e(0) \Phi'_a(0), \\ \lambda((f_{a,t})_z(0)) &= \lambda(f_z(0)) + t \lambda(e(0) \Phi'_a(0)) = k. \end{aligned} \quad (63)$$

Indeed, the ambient perturbation at $z = 0$ equals $\tilde{f}(0)$ for every t , so the chain rule always evaluates $d\tilde{\pi}_X$ at the same point $\tilde{f}(0)$. Moreover, the product rule gives $(\Xi_{\rho_\varepsilon})'_a(0) = \theta(0) \Phi'_a(0)$, so radial

regularization does not alter the derivative of the increment at the origin. Formula (63) fixes the displayed scalar component, not the entire first jet. For this choice, part (iii) and the preceding trace bound give $\sum_a (H_{a,1} + H_{a,i}) \leq -\kappa\delta$. One of the $2n$ real axes therefore has second area derivative $H_{a,\tau} \leq -\kappa\delta/(2n)$. Choose the sign of s so that $\ell_{a,\tau}s \leq 0$. Then (52) gives

$$\text{Area}_R(f_{a,\tau s}) - \text{Area}_R(f) \leq -\frac{\kappa\delta}{4n}s^2 + C_T|s|^3.$$

Before selecting the step, decrease the engine parameter radius to $\min\{t_0, 1, \epsilon/(2\max\{1, C_{\theta,\infty}\})\}$ if necessary, so that the selected perturbation segment lies in the compact tube used in Proposition 4.10. Set

$$t_* := \min\left\{\frac{t_0}{2}, \frac{\kappa\delta}{8n\max\{C_T, 1\}}\right\}, \quad \sigma := \frac{\kappa\delta}{8n}t_*^2. \quad (64)$$

Take $s \in \{t_*, -t_*\}$ with the chosen sign and set $g = f_{a,\tau s}$. The Taylor estimate gives $\text{Area}_R(g) \leq \text{Area}_R(f) - \sigma$. The selected increment has the bounds in (54). Proposition 4.10 gives the remaining estimate in (7) with

$$\beta_I = C_{A,I}t_*/\pi,$$

where $C_{A,I}$ is (57) for this increment. The parameter choice keeps its entire segment in the common tube. The constants $t_0, C_T, C_{A,I}$ depend only on the fixed caps and target data. The multiplier has the same cost for every nonzero covector $\lambda_{x_0}^F$ and every scalar k . Thus t_*, σ, β_I are independent of the radius, the disk, the center in F , and the scalar. $\square$

For scalar enlargement, it suffices to lift one finite-area disk and perturb its lift in one constant ambient direction.

Proof of Lemma 2.2. Choose $s_j \uparrow R$. Each restriction $h|_{\mathbb{D}_{s_j}}$ extends holomorphically past its closed disk and has the common bounds

$$\text{Area}_{s_j}(h) \leq A_0, \quad B_{s_j}(h) \leq B_0.$$

Apply Theorem 4.7(ii) with these caps. It supplies one compact set $K = K_{A_0, B_0} \Subset \mathcal{Y}$, one ambient neighborhood $\Omega \subset \mathbb{C}^{2m}$ of K , and one holomorphic projection $\Pi = \tilde{\pi}_X : \Omega \rightarrow X$ satisfying $\Pi|_{\mathcal{Y} \cap \Omega} = \pi_X$. The corresponding lifts H_j obey

$$H_j(\overline{\mathbb{D}_{s_j}}) \subset K, \quad \Pi \circ H_j = h \quad \text{on } \mathbb{D}_{s_j}, \quad \int_{\mathbb{D}_{s_j}} |H'_j|^2 dA \leq D_{\tilde{f}},$$

where the last constant is independent of j .

The compact range makes the coordinate functions of H_j a normal family on every compact subdisk of $\mathbb{D}_R$. A diagonal subsequence therefore converges locally uniformly, together with its derivatives, to a holomorphic map $H : \mathbb{D}_R \rightarrow \mathbb{C}^{2m}$. Since K is closed, $H(\mathbb{D}_R) \subset K$, and continuity gives $\Pi \circ H = h$. For every $s < R$, convergence of derivatives on $\overline{\mathbb{D}_s}$ gives $\int_{\mathbb{D}_s} |H'|^2 dA \leq D_{\tilde{f}}$. Exhausting $\mathbb{D}_R$ yields

$$H(\mathbb{D}_R) \subset K, \quad \Pi \circ H = h, \quad \int_{\mathbb{D}_R} |H'|^2 dA \leq D_{\tilde{f}}. \quad (65)$$

No compatibility between the separately chosen lifts is required.

Enlarge $D = D_{\tilde{f}}$ to at least 1 and put

$$Q = \sqrt{D/\pi}, \quad \bar{t} = \min\{1, \epsilon/(2\max\{1, Q\})\}.$$

Here ϵ is the common tube radius used in Section 4.3. For $v = H'(0)$, Lemma 4.9 gives

$$R|v| \leq Q, \quad \|zv\|_\infty \leq Q, \quad \int_{\mathbb{D}_R} |(zv)'|^2 dA = \pi R^2 |v|^2 \leq D.$$

Thus the affine formula (8) stays in the common tube. Its derivative is $(h_t)_z(0) = (1+t)h_z(0)$. Proposition 4.10, valid on the open disk by the same pointwise proof and exhaustion, gives the area and B^2 bounds with

$$C_{II} = D(L_g Q + 3M_0). \quad (66)$$

Source dilation satisfies the exact identities

$$\text{Area}_{(1+t)R}(G_t) = \text{Area}_R(h_t), \quad B_{(1+t)R}(G_t) = B_R(h_t).$$

It also restores the entire derivative at zero. The displacement estimate refers to corresponding source points:

$$\text{dist}_\omega(G_t((1+t)z), h(z)) \leq \sqrt{M_0} Q t.$$

After restricting to $\mathbb{D}_{R'}$, this comparison applies only where $|(1+t)z| < R'$. No closeness estimate for a principal recovery follows from this finite deformation bound. This proves the full-domain assertions. If h extends past the closed disk, its lift can be chosen with that property directly from Theorem 4.7, so both families have the corresponding extension. For an open disk, restrict at $R' = (1+t/2)R$. This lies strictly inside the new open domain. Restriction decreases both area and B , proving (12). $\square$

4.5. Scalar repair by an absolute radius increment

A second explicit enlargement repairs only the scalar derivative. Fix the same area and derivative caps A_0, B_0 as in the two operations. Take $f \in \mathcal{A}_r^{F, \lambda^F}$ with $\text{Area}_r(f) \leq A_0$ and $B_r(f) \leq B_0$; in particular, f extends holomorphically past $\overline{\mathbb{D}}_r$ and satisfies the scalar normalization. Set $x = f(0)$ and $\lambda = \lambda_x^F$. Choose a positive increment Δ . Put $R = r + \Delta$ and $\alpha = r/R$. Start on the larger disk with $f^{(\alpha)}(z) = f(\alpha z)$. Then

$$\text{Area}_R(f^{(\alpha)}) = \text{Area}_r(f), \quad B_R(f^{(\alpha)}) = B_r(f), \quad \lambda((f^{(\alpha)})_z(0)) = \alpha.$$

Choose the compatible lift and frame for this rescaled map. Set

$$L_f = \lambda \circ e(0), \quad c_f = \frac{L_f^*}{L_f L_f^*}, \quad \lambda_{\min} = \min_{y \in F} \|\lambda_y^F\|_{\omega, *}, \quad c_0 = \max\{1, \sqrt{\Gamma}/\lambda_{\min}\}.$$

The minimum is positive by compactness of F . The frame bound therefore gives $|c_f| \leq c_0$ and $L_f c_f = 1$, with c_0 uniform in the center. Thus

$$\hat{\phi}(z) = \frac{z}{R} \frac{c_f}{c_0}, \quad \hat{\Xi}(z) = \theta(\rho z) \hat{\phi}(z)$$

has uniform supremum and Dirichlet bounds; any fixed $0 < \rho < 1$ is allowed. Here the engine is applied with $D_\Phi = \pi$ and $N_\Phi = 1$, so denote its parameter radius by $\hat{t}_{0, \text{abs}}$. With $S_{\text{abs}} = C_{\theta, \infty}$, set

$$t_{0, \text{abs}} = \min\{\hat{t}_{0, \text{abs}}, 1, \epsilon/(2 \max\{1, S_{\text{abs}}\})\}.$$

This parameter is chosen separately from the descent parameter and ensures the same strict tube margin. The family $\hat{f}_t = \Pi(H + t\hat{\Xi})$ satisfies

$$\hat{f}_t(0) = x, \quad \lambda((\hat{f}_t)_z(0)) = \alpha + \frac{t}{c_0 R}.$$

The explicit choice is

$$t = c_0 R(1 - \alpha) = c_0 \Delta. \quad (67)$$

Provided $c_0 \Delta \leq t_{0, \text{abs}}/2$, the new disk $g = \hat{f}_{c_0 \Delta}$ is admissible on the larger closed disk. Proposition 4.10 supplies a constant $C_{A, \text{abs}}$ independent of r, R such that

$$\begin{aligned} R_{\text{new}} &= r + \Delta, \\ |\text{Area}_R(g) - \text{Area}_r(f)| &\leq c_0 C_{A, \text{abs}} \Delta, \\ |B_R(g)^2 - B_r(f)^2| &\leq c_0 C_{A, \text{abs}} \Delta / \pi. \end{aligned} \quad (68)$$

The coefficient in (68) is independent of both radii. This variant may replace the affine enlargement in the finite stages by choosing a common positive Δ with area cost at most η_0 ; the radius counter then uses $\lceil(\ell_F M - R_s)/\Delta\rceil$. The affine formula has the additional property of preserving the entire first jet.

5. COMPACTNESS, MARKED VALUES, AND FINITE-AREA DISKS

The main proof uses compactness to obtain a principal map in the original source coordinate and a rational chain retaining two marked values. We obtain both from one fixed-mark exhaustion statement, with a joint area bound for the principal map and the attached spheres. Point selection then gives the weighted derivative bound for an individual finite-area disk. Throughout this section, (X, ω) is compact Kähler; a Ricci lower bound is not required.

5.1. Finite-area removal

Finite area fills the punctures of a principal limit, including infinity when its source is $\mathbb{C}$. We first prove removal so that the compactness statement can include the extended principal map.

With $g_\omega(v, w) = \omega(v, Jw)$, the energy convention in the compactness and removal results used below is

$$\|df\|_{L^2}^2 = \int (|f_x|_{g_\omega}^2 + |f_y|_{g_\omega}^2) dA = 2 \int f^* \omega = 2 \text{Area}(f), \quad (69)$$

because $f_y = Jf_x$. Thus their energy hypotheses are equivalent to our area hypotheses, with the fixed factor 2.

Proof of Theorem 2.4. Finite area gives $\int_{0 < |z| < r} f^* \omega \rightarrow 0$ as $r \downarrow 0$, so every sufficiently small concentric annulus has small energy by (69). Compactness of X supplies completeness of the target metric and uniform continuity of its complex structure. The removal theorem of Ivashkovich and Shevchishin [29, Corollary 3.6] therefore gives a continuous extension at the puncture. In a target chart about the limiting value, the coordinate functions extend holomorphically by the Riemann removable singularity theorem. For an entire map, $\zeta = 1/z$ identifies infinity with a puncture and preserves area, so the same argument applies there. See also [37, Theorem 4.2.1]. $\square$

5.2. Marked compactness on an exhaustion

Spheres obtained by rescaling near energy concentration points go back to Sacks and Uhlenbeck [43, pp. 2–3 and Section 4]; Gromov’s compactness theory records how such components attach [24]. We use the theorem of Ivashkovich and Shevchishin [30]. For precise locators, we refer to its preprint formulation, whose title and numbering differ from the published version: Definition 1.6, Theorem 1.1, and the marked-point construction in Remark 2 following that theorem [29].

The *principal map* is the limit in the original source coordinate and may be constant. A *bubble* is a sphere obtained by rescaling. A *ghost sphere* is a constant component retaining marks or separating nodes; it has at least three special points, namely marks or nodes. The fixed-coordinate root disk has no such stability requirement. The dual graph has a vertex for each component and an edge for each node; here it is a rooted tree. The finite-area extension of an entire principal map is also a sphere and need not be a bubble.

The following proposition combines the principal limit used for recovery with the marked information needed for incidence. Fixed distinct source marks suffice for both applications. Constant components inside concentration trees must still be retained, since they can carry a marked value or join nonconstant components.

Proposition 5.1. *Let $R_j > 0$ and let $f_j : U_j \rightarrow X$ be holomorphic, where $U_j \subset \mathbb{C}$ is open and $\mathbb{D}_{R_j} \Subset U_j$. Suppose $\sup_j \text{Area}_{R_j}(f_j) < \infty$, and set $A_\infty = \liminf_j \text{Area}_{R_j}(f_j)$. Choose $R_\infty \in (0, \infty]$ with $R_\infty \leq \liminf_j R_j$, interpreting $\mathbb{D}_\infty$ as $\mathbb{C}$. Fix finitely many pairwise distinct points $a_1, \dots, a_m \in$*

$\mathbb{D}_{R_\infty}$, allowing $m = 0$. After first passing to a subsequence realizing A_∞ and then to a further subsequence, the following data exist simultaneously.

- (i) A finite set $\Sigma_\infty \subset \mathbb{D}_{R_\infty}$ and a holomorphic principal map $f_\infty : \mathbb{D}_{R_\infty} \rightarrow X$ satisfy

$$f_j \longrightarrow f_\infty \quad \text{smoothly on compact subsets of } \mathbb{D}_{R_\infty} \setminus \Sigma_\infty$$

in the original source coordinate.

- (ii) Finitely many holomorphic sphere components $u_\alpha : \mathbb{P}^1 \rightarrow X$, possibly constant, form trees attached to the principal root at points of Σ_∞ . Each attached tree contains a nonconstant component; each constant sphere is stable. If N_α is the set of nodes on the component and z_α is the root attachment point of the tree containing that component, there are source reparametrizations, defined near each compact subset of $\mathbb{P}^1 \setminus N_\alpha$ for all large j , with

$$f_j \circ \varphi_{j,\alpha} \longrightarrow u_\alpha, \quad \varphi_{j,\alpha} \longrightarrow z_\alpha$$

smoothly for the first convergence and locally uniformly for the second, on compact subsets of $\mathbb{P}^1 \setminus N_\alpha$.

- (iii) Root, spheres, nodes and marks form one nodal map. Writing u_v for its component maps, p_{vw}, p_{wv} for the two points of a node, and p_ℓ on v_ℓ for the limiting ℓ th mark, one has

$$u_v(p_{vw}) = u_w(p_{wv}), \quad f_j(a_\ell) \longrightarrow u_{v_\ell}(p_\ell).$$

A marked value therefore propagates along any path of constant components to its first nonconstant component.

- (iv) The areas, counted with the multiplicities of the maps, satisfy

$$\text{Area}_{R_\infty}(f_\infty) + \sum_\alpha \text{Area}(u_\alpha) \leq A_\infty. \quad (70)$$

Proof. Retain a subsequence whose total areas converge to A_∞ . The proof first selects the principal map on the whole exhaustion, then makes one marked extraction on a fixed disk, and finally counts their areas together.

The original-coordinate principal map.

A diagonal weak selection of the area measures on compact subdisks gives a Radon measure μ on $\mathbb{D}_{R_\infty}$. For every continuous compactly supported χ with $0 \leq \chi \leq 1$,

$$\int \chi d\mu = \lim_j \int \chi f_j^* \omega \leq A_\infty,$$

so $\mu(\mathbb{D}_{R_\infty}) \leq A_\infty$ by exhaustion. Fix $p > 2$ and an area threshold $\epsilon_0 > 0$ small enough for the local compactness result [29, Lemma 3.1 and Corollary 3.3], using (69), and set

$$\Sigma_\infty = \{z : \mu(\{z\}) \geq \epsilon_0\}.$$

The mass bound makes this set finite. For $z \notin \Sigma_\infty$, choose a source disk with μ -null boundary and μ -mass less than ϵ_0 . Weak convergence puts its area below that threshold for all large j . The cited result gives strong local $W^{1,p}$ convergence on a smaller disk; Sobolev embedding gives uniform convergence, and target charts and Cauchy estimates give smooth convergence. A diagonal choice on a countable cover gives f_∞ on the complement of Σ_∞ in the original coordinate. Its local definitions agree because they are limits of the same subsequence. Smooth convergence and exhaustion give $\text{Area}(f_\infty) \leq A_\infty$. Theorem 2.4 fills the finitely many punctures without changing area.

One marked extraction and its root.

Choose fixed radii $0 < r < \rho < r' < R_\infty$ such that $\Sigma_\infty \cup \{a_1, \dots, a_m\} \subset \mathbb{D}_r$. Apply the marked compactness theorem on $\mathbb{D}_{r'}$, retaining $\mathbb{D}_\rho$, to the subsequence already chosen. The standard complex structure is fixed on this source disk, so the boundary-annulus nondegeneration condition is automatic [29, Definition 1.7 and Theorem 1.1]. The positive collar between $\mathbb{D}_\rho$ and $\partial\mathbb{D}_{r'}$ keeps

this application strictly inside the available domains. The target is compact, and the energy bound follows from (69).

The fixed-domain construction uses the original coordinate away from concentration points and creates sphere components by rescaling shrinking regions at those points [29, proof of Theorem 1.1, Case 1, and the remark following its proof]. Its root coordinate is identified with the original disk: the holomorphic identification agrees with the identity on an outer open set and hence everywhere. Uniqueness of the already selected limit then identifies the root map with f_∞ off Σ_∞ , and holomorphic extension identifies it throughout $\mathbb{D}_\rho$. We extend this root by the global map f_∞ .

The marked construction in Remark 2 following the same theorem retains the marked evaluations, inserting stable constant components when a mark approaches a node in a rescaled coordinate. Nonconstant components can attach only in Σ_∞ , since convergence is smooth elsewhere. Nor can a tree made entirely of constant components attach to the root: if it had $g \geq 1$ vertices and m_z marks, stability would give

$$3g \leq 2(g - 1) + 1 + m_z, \quad m_z \geq g + 1 \geq 2.$$

All these marks would have the same original-coordinate limit, contrary to the fixed distinct marks. Thus all attached trees lie over Σ_∞ and contain a nonconstant sphere. The shrinking-region construction gives the reparametrizations and their collapse asserted in (ii), including for internal ghosts. The compactness theorem gives finiteness of the marked tree.

By Definition 1.6 of the cited preprint, all components belong to one continuous nodal limit. Its two branches therefore have equal values at every node. Marked convergence gives the second identity in (iii); repeatedly applying the node identity propagates a marked value through constant components. No derivative at a marked point is asserted to survive.

Joint global area accounting.

Choose a compact subset $Q \subset \mathbb{D}_{R_\infty} \setminus \Sigma_\infty$ and compact subsets $Q_\alpha \subset \mathbb{P}^1 \setminus N_\alpha$ on the sphere components. The fixed extraction represents the Q_α in mutually disjoint shrinking source regions at separated scales. For all large j these regions also avoid Q , even when Q extends outside the retained disk $\mathbb{D}_\rho$, because Q stays a positive distance from Σ_∞ . Taking the sequence limit first gives

$$\int_Q f_\infty^* \omega + \sum_\alpha \int_{Q_\alpha} u_\alpha^* \omega = \lim_j \left(\int_Q f_j^* \omega + \sum_\alpha \int_{\varphi_{j,\alpha}(Q_\alpha)} f_j^* \omega \right) \leq A_\infty.$$

Now exhaust the global principal domain minus Σ_∞ and each sphere minus its nodes. The omitted points have zero area, and all omitted source areas are nonnegative. This proves (70). Constant components contribute zero; multiplicities are retained because we integrate pullbacks by the component maps. $\square$

Taking no marks gives exactly the principal compactness statement used in the continuation argument.

Proof of Proposition 2.3. Apply Proposition 5.1 with $m = 0$ and omit the nonnegative sphere areas in (70). Its principal map and convergence have the required properties. $\square$

5.3. Two marked values and incidence

Bubbling at 0 preserves the marked value but can separate a prescribed tangent direction from the component carrying the area. Fixing even the complete first jet at 0 does not prevent this loss. On the unit disk, with the product Fubini–Study metric, consider

$$\widehat{f}_j(z) := ([jz^2 : 1], [z : 1]) : \mathbb{D} \longrightarrow \mathbb{P}^1 \times \mathbb{P}^1.$$

The areas are uniformly bounded, and in the affine coordinates $([w : 1], [v : 1])$ the complete first jet is fixed:

$$\widehat{f}_j(0) = ([0 : 1], [0 : 1]), \quad (\widehat{f}_j)_z(0) = (0, 1).$$

Away from zero, the principal limit is

$$\widehat{f}_\infty(z) = ([1 : 0], [z : 1]).$$

Its extension has value $([1 : 0], [0 : 1])$ at zero, different from the marked value. Rescaling by $z = \zeta/\sqrt{j}$ instead gives

$$\widehat{f}_j(\zeta/\sqrt{j}) = ([\zeta^2 : 1], [\zeta/\sqrt{j} : 1]) \longrightarrow ([\zeta^2 : 1], [0 : 1]).$$

The prescribed second-factor direction remains on the residual principal component. Correspondingly, $B_1(\widehat{f}_j)/|(\widehat{f}_j)_z(0)|_\omega \rightarrow \infty$. Thus prescribing the first jet at 0 does not ensure that a nonconstant bubble retains that tangent direction.

Marked and nodal values retain incidence. Lemma 2.5 gives a chain joining two compact target sets with controlled total area. It supplies the compactness alternative in Section 2 and the final passage to the sharp area bound.

Proof of Lemma 2.5. Pass first to a subsequence realizing the original lower limit $A_\infty := \liminf_j \text{Area}_{R_j}(f_j)$. Apply Proposition 5.1 with $R_\infty = \infty$ and the fixed marks 0, 1. It supplies an entire principal map f_∞ and one finite marked tree, with the joint area bound (70). Theorem 2.4 fills infinity, producing the principal sphere. Replace the principal root of the marked limit by this extension and retain its attached sphere trees. Each mark lies either on the principal sphere or on one of these trees. Its marked value lies in K_0 or K_1 because these sets are closed. On the resulting finite tree, take the path between the marked components and omit constant components. Nodal equality gives nonconstant sphere maps $u_1, \dots, u_\ell$ with

$$u_1(\mathbb{P}^1) \cap K_0 \neq \emptyset, \quad u_\ell(\mathbb{P}^1) \cap K_1 \neq \emptyset, \quad u_i(\mathbb{P}^1) \cap u_{i+1}(\mathbb{P}^1) \neq \emptyset \quad (1 \leq i < \ell).$$

There is at least one such component since $K_0 \cap K_1 = \emptyset$. The components on the path are selected from the global principal sphere and the sphere components of the fixed rooted limit. The joint area estimate (70), after omitting unused components of nonnegative area, gives

$$\sum_{i=1}^{\ell} \int_{\mathbb{P}^1} u_i^* \omega \leq A_\infty.$$

The path belongs to the single marked limit supplied by the proposition. Components arising near the moving outer boundary are not needed for its incidence or area estimate. $\square$

5.4. Rescaling and finite-area weighted derivatives

Brody reparametrization [5, Lemma 2.1], in the affine form described by Duval [16, Section 1], turns divergence of B_R into a nonconstant entire limit. For a single finite-area disk, the positive rescaled area cannot persist in shrinking boundary collars; this gives Lemma 2.7.

Lemma 5.2. *Let $f_j : U_j \rightarrow X$ be holomorphic maps, where $U_j \subset \mathbb{C}$ is open and $\overline{\mathbb{D}}_{R_j} \Subset U_j$, and suppose $B_{R_j}(f_j) \rightarrow \infty$. Then after affine source rescaling, a subsequence converges locally smoothly on $\mathbb{C}$ to a nonconstant entire map $u : \mathbb{C} \rightarrow X$. If in addition $\sup_j \text{Area}_{R_j}(f_j) < \infty$, then*

$$\text{Area}(u) \leq \liminf_j \text{Area}_{R_j}(f_j).$$

Proof. Under the additional area hypothesis, first pass to a subsequence for which $\text{Area}_{R_j}(f_j)$ converges to the lower limit of the original area sequence. In either case, choose w_j maximizing $(R_j - |z|)|(\widehat{f}_j)_z(z)|$, and set $d_j := R_j - |w_j|$, $\Lambda_j := |(\widehat{f}_j)_z(w_j)|$. Then

$$d_j \Lambda_j = B_{R_j}(f_j) \longrightarrow \infty, \quad |(\widehat{f}_j)_z(z)| \leq 2\Lambda_j \quad (|z - w_j| < d_j/2).$$

Indeed, $|z - w_j| < d_j/2$ implies $R_j - |z| > d_j/2$. Maximality then gives $(d_j/2)|(\widehat{f}_j)_z(z)| \leq d_j \Lambda_j$, which is the second bound. The rescaled map

$$u_j(\zeta) := f_j(w_j + \zeta/\Lambda_j), \quad |\zeta| < \frac{1}{2} d_j \Lambda_j,$$

satisfies $|(u_j)_\zeta|_\omega \leq 2$ and $|(u_j)_\zeta(0)|_\omega = 1$. The source radii $d_j \Lambda_j/2$ tend to infinity. On every fixed compact source disk, the derivative bound gives equicontinuity, and compactness of X gives a uniformly convergent subsequence. A diagonal selection and Cauchy estimates in local target charts give a smooth limit $u : \mathbb{C} \rightarrow X$ with $|u_\zeta(0)|_\omega = 1$; hence u is nonconstant. Under the additional area hypothesis, conformal invariance gives, for each fixed $S > 0$,

$$\text{Area}_S(u) = \lim_j \text{Area}_S(u_j) \leq \lim_j \text{Area}_{R_j}(f_j) = \liminf_j \text{Area}_{R_j}(f_j).$$

Letting $S \rightarrow \infty$ proves the area bound. $\square$

Proof of Lemma 2.7. Suppose $B_R(h) = \infty$. As $s \uparrow R$, the quantities $B_s(h)$ increase to $B_R(h)$: for each fixed $z \in \mathbb{D}_R$, $(s - |z|)|h_z(z)|_\omega$ tends to $(R - |z|)|h_z(z)|_\omega$ once $s > |z|$, and the suprema increase. Choose $s_j \uparrow R$ with $B_{s_j}(h) \rightarrow \infty$, and apply the point-selection construction of Lemma 5.2 to $h|_{\mathbb{D}_{s_j}}$. These restrictions extend holomorphically past their closed disks. The construction gives centers $w_j \in \mathbb{D}_{s_j}$ and numbers $\Lambda_j = |h_z(w_j)|_\omega$ such that

$$(s_j - |w_j|)\Lambda_j \rightarrow \infty, \quad u_j(\zeta) := h(w_j + \zeta/\Lambda_j) \rightarrow u(\zeta)$$

smoothly on compact subsets of $\mathbb{C}$, where $|u_\zeta(0)|_\omega = 1$. Because $s_j \leq R$, we have $\Lambda_j \rightarrow \infty$. The derivative of the fixed map h is bounded on each compact subdisk, so $|w_j| \rightarrow R$.

Choose $S > 0$ with $\text{Area}_S(u) > 0$. For all large j , the disk $\mathbb{D}_{S/\Lambda_j}(w_j)$ lies in $\mathbb{D}_{s_j}$, and conformal invariance and smooth convergence give

$$\int_{\mathbb{D}_{S/\Lambda_j}(w_j)} h^* \omega = \text{Area}_S(u_j) \rightarrow \text{Area}_S(u) > 0.$$

On the other hand, $|w_j| \rightarrow R$ and $S/\Lambda_j \rightarrow 0$ place these disks in arbitrarily thin boundary collars of $\mathbb{D}_R$. Since $\text{Area}_R(h) < \infty$, $\int_{\mathbb{D}_R \setminus \overline{\mathbb{D}_s}} h^* \omega \rightarrow 0$ as $s \uparrow R$. This contradicts the preceding positive limit. $\square$

6. A FINITE CATALOGUE AND A FINITE-LOOP ALTERNATIVE

There is a second organization of recovery. It replaces a high- B disk by a member of a finite catalogue, keeping its scalar derivative and allowing its center to move in F . The catalogue is prepared by compactness and a finite-cover selection. Once it has been selected, the subsequent iteration uses only finitely many operations for each scalar-derivative target. These finite loops give an alternative to taking successive limits of recovered maps as in Section 2.3.

Fix a source radius $R_0 > 0$, an area cap C , and a real scalar window $0 < k_- \leq k \leq k_+ < \infty$. The disks in this section satisfy

$$f(0) \in F, \quad \lambda_{f(0)}^F(f_z(0)) = k. \quad (71)$$

The real positive window suffices for the iteration below.

Proposition 6.1. *One can select either a sphere meeting and escaping F of area at most C , or a finite collection of holomorphic maps G_i , numbers $k_i > 0$, parameter neighborhoods $W_i \subset [k_-, k_+] \times [0, C]$, and a number $M < \infty$ with the following property. Every map f holomorphic near $\overline{\mathbb{D}_{R_0}}$, satisfying (71), $\text{Area}_{R_0}(f) \leq C$, and $B_{R_0}(f) > M$, has $(k, \text{Area}_{R_0}(f)) \in W_i$ for at least one i . Select the first such index and define*

$$g(z) = G_i((k/k_i)z), \quad z \in \mathbb{D}_{R_0}. \quad (72)$$

This map extends across a neighborhood of the closed source disk, satisfies (71) with the same k , and obeys

$$\text{Area}_{R_0}(g) \leq \text{Area}_{R_0}(f) - e_X/2, \quad R_0 k / \ell_F \leq B_{R_0}(g) \leq M. \quad (73)$$

The source extension neighborhood can be chosen uniformly for the members of each catalogue family in (72).

Proof. Let $Z = [k_-, k_+] \times [0, C]$. For positive integers m , let K_m be the closure in Z of the parameter pairs $(k, \text{Area}_{R_0}(f))$ of admissible disks with $B_{R_0}(f) > m$. These are decreasing compact sets. Put $P = \bigcap_{m \geq 1} K_m$.

For $p = (k_p, E_p) \in P$, choose such disks f_j with $B_{R_0}(f_j) > j$ and parameter pairs tending to p . Apply relative compactness and the area-loss argument of Lemma 2.8. These apply to the varying scalar: the maps $z \mapsto f_j(z/k_j)$ have scalar 1, radius $k_j R_0$ tending to $k_p R_0 > 0$, and unchanged area and B . Rescaling the principal map back by $z \mapsto k_p z$ gives an open-disk map $h_p : \mathbb{D}_{R_0} \rightarrow X$ with scalar k_p , center in F , and

$$\text{Area}_{R_0}(h_p) \leq E_p - e_X.$$

If the alternative supplies an escaping sphere at any such choice, retain it as the first result of this proposition.

For a principal result use the affine enlargement with a sufficiently small positive parameter. Its normalized source dilation preserves the full derivative, including its scalar k_p . By Lemma 2.2 the enlargement costs less than $e_X/8$. Restrict at a radius $L_p > R_0$ strictly inside the enlarged open disk, obtaining a map G_p holomorphic near $\overline{\mathbb{D}}_{L_p}$, with scalar k_p and

$$\text{Area}_{L_p}(G_p) \leq E_p - 7e_X/8.$$

The same lemma and its proof work for a nonunit scalar, since the affine enlargement preserves the entire derivative after dilation.

Choose $R_0 < L'_p < L_p$. In a sufficiently small relative open neighborhood W_p of p in Z , every (k', a') satisfies

$$(k'/k_p)R_0 < L'_p, \quad |a' - E_p| < e_X/8.$$

The maps $G_p((k'/k_p)z)$ are then defined on one common neighborhood of $\overline{\mathbb{D}}_{R_0}$. For example, shrink W_p again to bound k'/k_p strictly below L'_p/R_0 . Their areas are at most $E_p - 7e_X/8 \leq a' - 3e_X/4$, and their weighted derivatives satisfy

$$B_{R_0}(G_p((k'/k_p) \cdot)) = B_{(k'/k_p)R_0}(G_p) \leq B_{L_p}(G_p) < \infty.$$

Their centers lie in F and their scalars are exactly k' .

Compactness of P selects finitely many of these neighborhoods, say $W_1, \dots, W_s$, covering P . If P is empty, take this collection to be empty. The sets

$$W_1, \dots, W_s, \quad Z \setminus K_1, Z \setminus K_2, \dots$$

form an open cover of Z . A finite subcover and the decreasing property of K_m give an integer m_0 with $K_{m_0} \subset \bigcup_{i=1}^s W_i$; if no complementary set was needed, take $m_0 = 1$. Choose

$$M \geq \max\{1, m_0, B_{L_1}(G_1), \dots, B_{L_s}(G_s)\}.$$

For every disk with $B > M$, its parameter pair belongs to K_{m_0} , so the first applicable catalogue map has all the stated properties, including the weaker decrease $e_X/2$ in (73). In the empty-catalogue case there are no such disks, and the assertion about replacement is vacuous. $\square$

The catalogue is chosen once for the fixed parameter rectangle. A later disk is replaced by rescaling the first applicable member; its center may change within F .

Proposition 6.2. *Fix $\delta > 0$ and a normalized closed coordinate seed on $\mathbb{D}_{R_0}$ with scalar 1 and area $A_s < A_*$. For every $K > 1$, the catalogue alternative followed by the following finite loop gives either an escaping sphere of area at most $C = A_* + 2\delta$, or a closed disk on $\mathbb{D}_{R_0}$ with scalar at least K and area less than C .*

Proof. Prepare Proposition 6.1 for the scalar window $[1, 2K]$ and cap C . Retain a sphere if it is supplied. Otherwise enlarge M to include the seed's B ; the same catalogue remains valid for this larger cutoff. Use this one B cap for both surgeries. Surgery I preserves every scalar k , not just 1:

the vanishing and kernel conditions on its multiplier are unchanged. Its decrease $\sigma > 0$ is therefore uniform throughout the window.

In Surgery II retain $h_t(z) = \Pi(H(z) + tzH'(0))$ on the *same* radius R_0 , without the source dilation. Its scalar becomes $(1+t)k$. Its area and B^2 changes obey the same estimates as in Section 4.4. Fix

$$\tau = \min\{\bar{t}/2, \eta_0/\max\{1, C_{II}\}\} > 0, \quad L = A_* + \delta, \quad \eta_0 = \delta/2.$$

In particular $\tau \leq 1$. At the start of each loop $B \leq M$. If $k \geq K$, stop. Otherwise do Surgery I when $A \geq L$, and the same-radius Surgery II with $t = \tau$ when $A < L$. If the resulting B exceeds M , immediately apply (72) once. This restores $B \leq M$, lowers area by at least $e_X/2$, and keeps the scalar just obtained.

Area stays below C at all times. While enlargement is needed the old scalar is less than K , so the new scalar is less than $2K$. All catalogue and surgery calls are therefore within their declared caps. After

$$N = \left\lceil \frac{\log K}{\log(1+\tau)} \right\rceil$$

enlargements, the scalar has reached K . For every completed prefix of the loop, with operation counts N_I, N_{II}, N_{III} , the exact inequalities just proved give

$$N_I\sigma + N_{III}e_X/2 \leq A_s + N_{II}\eta_0, \quad N_{II} \leq N. \quad (74)$$

Thus one can run a bounded loop with at most

$$N + \left\lceil \frac{A_s + N\eta_0}{\min\{\sigma, e_X/2\}} \right\rceil$$

individual surgery and catalogue operations. The stated scalar test has terminated it by that bound. All tests concern the actual current map, and a possible crossing of the B cap is repaired before the next surgery using that cap. $\square$

To turn these scalar targets into the relative sphere, choose a positive upper bound H for $|\eta \circ \psi|$ on V . For a disk with scalar k , the Cauchy estimate on every source subdisk contained in V is $k \leq H/r$. Hence when $2H/k < R_0$ there is a first-exit point z with

$$0 < |z| \leq 2H/k, \quad f(z) \in \partial V.$$

Indeed a disk of radius $2H/k$ whose entire image lies in V would have derivative at most $k/2$; restriction to the first exit along a radial segment supplies the stated point. Applying Proposition 6.2 with $K \rightarrow \infty$ therefore gives, unless a sphere has already been returned, maps $\zeta \mapsto f(z\zeta)$ on radii tending to infinity, with source marks $0, 1$ mapped into the disjoint compact sets $F, \partial V$. Their areas stay at most C . Select a subsequence whose areas converge. The two-marked chain theorem yields the required escaping sphere, and the last paragraph of the proof of Theorem 1.1 gives the sharp bound.

The retained transverse information is the base scalar and the two separated marked values. A positive amount of total area inside V alone could belong to a component contained in F . In the principal branch it is convergence of the bounded base compositions, and thus of their nonzero derivatives, that retains transverse information. The catalogue loop supplies this normalization explicitly; it does not rely on a localization conclusion from unmarked total area alone.

7. RATIONAL CONNECTEDNESS AND AMPLE TANGENT BUNDLES

The analytic construction supplies the curves needed for two classical algebraic arguments. We recall only the conditions that allow those arguments to apply; for the general theory of rational curves, see Kollár [34] and Debarre [11].

Proof of Theorem 1.3. Choose a positive-Ricci metric as in Proposition 3.1. Let $q : X \dashrightarrow Y$ be any quasifibration with $\dim Y > 0$, represented on dense open subsets by a proper morphism $q : X^\circ \rightarrow Y^\circ$ with connected fibers. After shrinking Y° , generic smoothness makes q smooth [28, III, Corollary 10.7]. For a general $y \in Y^\circ$, the full fiber $F = q^{-1}(y)$ is therefore smooth and compact. Theorem 1.1, applied to q near F , gives a rational curve meeting F but not contained in it. By Campana’s criterion [6, Proposition 2.7], this property for every such quasifibration implies rational chain connectedness. Here Campana’s term “rationnellement connexe” refers to rational chains. The characteristic-zero smoothing theorem then gives rational connectedness in the sense used here [2]. $\square$

Proof of Theorem 1.4. For $n = 1$, ampleness gives $\deg T_X = 2 - 2g(X) > 0$, so $X \simeq \mathbb{P}^1$. Assume $n \geq 2$. Since T_X is ample, its determinant $-K_X$ is ample [26]. Thus X is Fano, and Theorem 1.2 supplies a rational curve. Starting with this curve, the two-point degree-reduction argument of Mori [39] gives a rational curve of anticanonical degree at most $n + 1$; we use precisely the first step in the proof of [12, Section 5, Theorem 3]. That step degenerates an already existing rational curve and decreases its positive integral degree. The genus-zero stable-map compactification [22, Theorem 1] gives a modern setting for this degeneration.

Let $f : \mathbb{P}^1 \rightarrow X$ be the normalization of the resulting curve. Ampleness gives $f^*T_X = \bigoplus_{i=1}^n \mathcal{O}(a_i)$ with every $a_i \geq 1$. The nonzero differential $\mathcal{O}(2) \rightarrow f^*T_X$ forces some $a_i \geq 2$. Consequently its anticanonical degree is at least $n + 1$, and equality gives

$$f^*T_X \simeq \mathcal{O}(2) \oplus \mathcal{O}(1)^{\oplus(n-1)}.$$

The same differential is then nowhere zero. These are the standard minimal-curve conditions in [12, Section 5, Corollary (a),(c)]. The remaining classification argument of Mori, starting from such a curve, identifies X with $\mathbb{P}^n$; see [12, Section 6]. $\square$

The degree reduction and classification used here take place over $\mathbb{C}$. In the classical proof, the positive-characteristic production of the first curve and the subsequent spreading out are the second and third steps of [12, Section 5, Theorem 3]; our analytic construction replaces those steps. Related numerical characterizations of projective space were developed by Cho, Miyaoka, and Shepherd-Barron [9], with a shorter proof by Kebekus [32].

8. OTHER APPROACHES AND THE GEOMETRY OF THE CONSTRUCTION

A shorter existence argument.

The existence of the relative sphere can also be obtained without following the detailed surgery iteration. An area-profile argument requires only uniform area descent under fixed area and derivative caps, and enlargement of one finite-area disk with arbitrarily small area cost. The qualitative consequences of Proposition 2.1 and Lemma 2.2 suffice; their precise steps and the intermediate B estimates are not needed for this argument.

For $A > A_*$, define

$$a(R) = \inf\{\text{Area}_R(f) : f \in \mathcal{A}_R^{F, \lambda^F}\}, \quad R_A = \sup\{R : a(R) < A\},$$

with $\inf \emptyset = +\infty$. Restriction makes a nondecreasing. Small coordinate disks and extension across a closed source disk show that $\{R : a(R) < A\}$ is a nonempty open interval. At a finite endpoint, choose $r_j \uparrow R_A$ and almost minimizers whose areas tend to $L = a(R_A^-) \leq A$. Relative compactness either supplies an escaping sphere or gives a normalized principal map h on $\mathbb{D}_{R_A}$ with area at most L . Since every restriction of h is admissible,

$$a(s) \leq \text{Area}_s(h) \quad (s < R_A), \quad \text{Area}_{R_A}(h) = L.$$

The equality follows by exhaustion. The single map h has finite $B_{R_A}(h)$. If $L > A_*$, uniform descent on restrictions close to R_A gives competitors of area at most $L - \sigma$, incompatible with $a(s) \rightarrow L$.

Thus $L \leq A_* < A$. Enlarging h at sufficiently small area cost, then restricting strictly inside the enlarged source, gives an admissible closed disk beyond R_A with area less than A . Consequently a finite endpoint supplies the sphere. If $R_A = \infty$, expanding-source compactness and finite-area removal supply it. In both cases the two-marked passage in the proof of Theorem 1.1 gives the sharp bound.

Why retain the explicit surgeries?

The additional analytic work in the main construction serves a geometric purpose. We deform specified disks: curvature selects an area-decreasing direction, enlargement restores the normalized derivative on a larger source, and recovery retains a principal map with an accounted area loss. The formulas explain how these operations act and what each one costs. In particular, they control the changes of area and weighted derivative, and (61) controls displacement at corresponding source points. The finite-stage counters and the recovery depth turn these local estimates into a construction that continues to a sphere. The base normalization, the projected-area bound (13), and the marked chain identify the geometric information retained at the limiting steps.

This is the reason for developing the two concrete surgeries despite their greater technical demands. The area profile selects almost minimizers from whole classes of disks; it gives the existence conclusion without describing their successive deformations. Following the disks themselves gives a more direct geometric account of the construction, together with estimates that the shorter argument does not record. The finite catalogue in Section 6 provides another organization of recovery, with a bounded number of subsequent operations for each prescribed scalar target.

Relation to Mori's method.

For the surrounding birational theory, see Kollár and Mori [36]; Debarre [10, Sections 2–3] discusses bend-and-break and rational quotients. In Mori's argument, Frobenius amplifies the anticanonical degree of a map while keeping the genus of its source fixed. Deformation theory produces a moving family, degeneration produces rational components, and spreading out transfers the curves to characteristic zero [39, 38]. The disk construction instead uses Ricci-guided holomorphic deformations and finite-area compactness over $\mathbb{C}$. Its applications then invoke classical rational-quotient and chain-smoothing results, together with characteristic-zero two-point degree reduction and minimal-curve classification. The analytic production of the initial curves uses no bend-and-break.

Classical effectiveness includes anticanonical degree bounds. The every-point bound $-K_X \cdot C \leq n + 1$, attributed to Kollár in [40, p. 749], gives the same pointwise metric area bound as Theorem 1.2. Our contribution concerns the explicit analytic operations and the geometric information available during their iteration. The hierarchy still uses countable repetitions and subsequence selections, and the finite catalogue is prepared by compactness. Thus the effectiveness established here is quantified control of the analytic construction; obtaining equations of the final rational curve by a finite-data numerical algorithm remains a further problem.

Numerical positivity.

The essential limitation is the positivity hypothesis. The trace identity (30), with the multiplier cost (33), gives

$$4\pi(n + 1) - 2 \int_{\mathbb{D}_R} f^* \text{Ric}(\omega).$$

We use the pointwise inequality $\text{Ric}(\omega) \geq \kappa\omega$ to make this expression negative whenever the area exceeds A_* , uniformly over the disks encountered in the construction. This is positivity of the Ricci form in every tangent direction, not merely positivity of scalar curvature. By contrast, the Miyaoka–Mori criterion starts from the numerical inequality $-K_X \cdot C > 0$ for one curve C and produces rational curves through its points [38, Theorem 5]. The recent optimal degree refinement of Jovinelly, Lehmann, and Riedl retains this numerical hypothesis and the Frobenius step [31,

Theorem 1.1 and its proof]. The degree of a closed curve controls one integrated curvature quantity; it gives no comparable lower bound along all the open disks and deformations used here.

This distinction also matters for the uniruledness criterion of Boucksom, Demailly, Păun, and Peternell [4]. Their cone duality produces a covering family (C_t) with $K_X \cdot C_t < 0$ when K_X is not pseudo-effective. The deduction of uniruledness then explicitly uses Miyaoka–Mori [38, Theorem 5]. Thus cone duality and the production of rational curves are separate steps. Our present argument does not replace the latter step under this numerical hypothesis; a single such curve need not itself belong to a covering family. Campana also emphasizes the role of positive characteristic in this numerical approach [7, Theorems 3.4–3.5 and the following paragraph].

Remark 8.1. Let X be a smooth complex projective variety, let $f : C \rightarrow X$ be a nonconstant map from a smooth connected projective curve, and let $p \in C$. Can one construct a rational curve through $f(p)$ by holomorphic disk deformations and finite-area compactness under the sole assumption

$$-K_X \cdot f_*[C] > 0?$$

The existence conclusion is supplied by Miyaoka–Mori [38]; the problem is to obtain it by an analytic construction in characteristic zero. The missing ingredient is a replacement for uniform Ricci descent derived from this numerical positivity, together with area control through continuation. Such a construction for a covering family would provide an analytic replacement for the curve-production step in the uniruledness argument of Boucksom, Demailly, Păun, and Peternell [4].

The numerical problem above is related to Yang’s proposed analytic construction of rational curves under RC-positive curvature [47]. Ou extended the uniruledness criterion to compact Kähler manifolds [42], with a subsequent analytic approach by Cao and Păun [8]. These proofs still use the projective criterion of Boucksom, Demailly, Păun, and Peternell [4], and hence the positive-characteristic methods of Miyaoka–Mori [38]; replacing this decisive step analytically remains a central challenge.

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AI USE DISCLOSURE

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